

Optimizing reinforcement of Auxetic Lattice: An ML based approach

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Indian Institute of Technology Kharagpur

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Of

Master of Technology

By

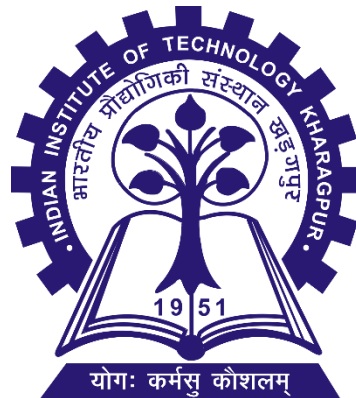
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CERTIFICATE

This is to certify that the thesis entitled **“Optimizing reinforcement of Auxetic Lattice: An ML based approach”**, submitted by **Anan Gupta**, in the Department of Mechanical Engineering, Indian Institute of Technology Kharagpur, India, for the award of the degree of Master of Technology, is a record of bona fide research work carried out by him under my supervision and guidance. The thesis fulfills all the requirements as per the regulations of this institute. Neither this thesis nor any part has been submitted for any degree or academic award elsewhere.

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1. Abstract

Auxetic materials, characterized by their negative Poisson's ratio, exhibit exceptional mechanical properties such as high energy absorption and fracture resistance. This work presents a novel methodology for optimizing the reinforcement of a re-entrant honeycomb (REH) lattice to achieve improved stiffness without significantly compromising auxetic behavior. The approach combines finite element (FE) simulations with deep learning-based modeling and gradient-based optimization. A data generation pipeline is developed in Abaqus to simulate various reinforcement configurations by embedding circular disc fibers at random locations within an 8×8 REH lattice. Each configuration is analyzed to extract its effective Young's modulus and Poisson's ratio. The resulting dataset, comprising reinforcement images, number of fibers, and their radii, is used to train a convolutional neural network (CNN) that predicts the mechanical properties directly from reinforcement patterns. Finally, a gradient descent-based optimization framework identifies the reinforcement configuration that maximizes stiffness while minimizing the Poisson's ratio. The proposed framework demonstrates the potential of integrating finite element analysis with machine learning for inverse design of auxetic metamaterials.

2. Introduction

2.1 Background

Low-density honeycomb structures are utilised in many fields such as civil engineering, aerospace engineering, chemical engineering, transportation, mechanical engineering, and sports. Honeycomb structures and re-entrant honeycomb (REH) structures have many advantages in terms of energy absorption, such as acoustic and thermal versatility, radio-frequency protection, air orientation, indentation resistance, synclastic behaviour, variable permeability, and favourable dielectric properties. In contrast to traditional engineering materials, materials with negative Poisson's ratios (NPRs) exhibit cross-sectional expansion when they are subjected to axial tensile load. As expected, if they are exposed to axial compression, their cross-sections exhibit contraction. Gibson et al. obtained the NPR property of cellular solids experimentally and theoretically for the deformed regular honeycomb. These types of materials are known as auxetic materials.[\[2\]](#)

Re-entrant structures are one of the most thoroughly studied of all auxetic structures. It stands out for its simplicity and scalability. The mechanical properties of the re-entrant structures are determined by the geometric properties of the repeating core cell. These parameters are the cell wall angle (θ), cell wall thickness (t), cell wall length (h), and inclined cell wall length (l) [\[2\]](#) as depicted in Fig. 2.1(a). For this study, an 8x8 re-entrant lattice would be studied as our Representative volume element (RVE) as depicted in Fig. 2.1(b).

2.2 Motivation

Studies by Choi and Lakes suggested that the fracture toughness is higher in the case of auxetic foams compared to the conventional foams even though the auxetic foams exhibit low Young's modulus [\[3\]](#). These attributes lead to auxetic materials having potential applications in different industries. Many researchers attempted to improve the in-plane elasticity of the structure by modifying the structure of a unit cell. Reinforcing these lattices at optimal locations is one such approach that can enhance stiffness without losing their auxetic nature. However, finding the best reinforcement pattern is a complex, high-dimensional optimization problem due to the non-linear dependence of mechanical properties on geometry.

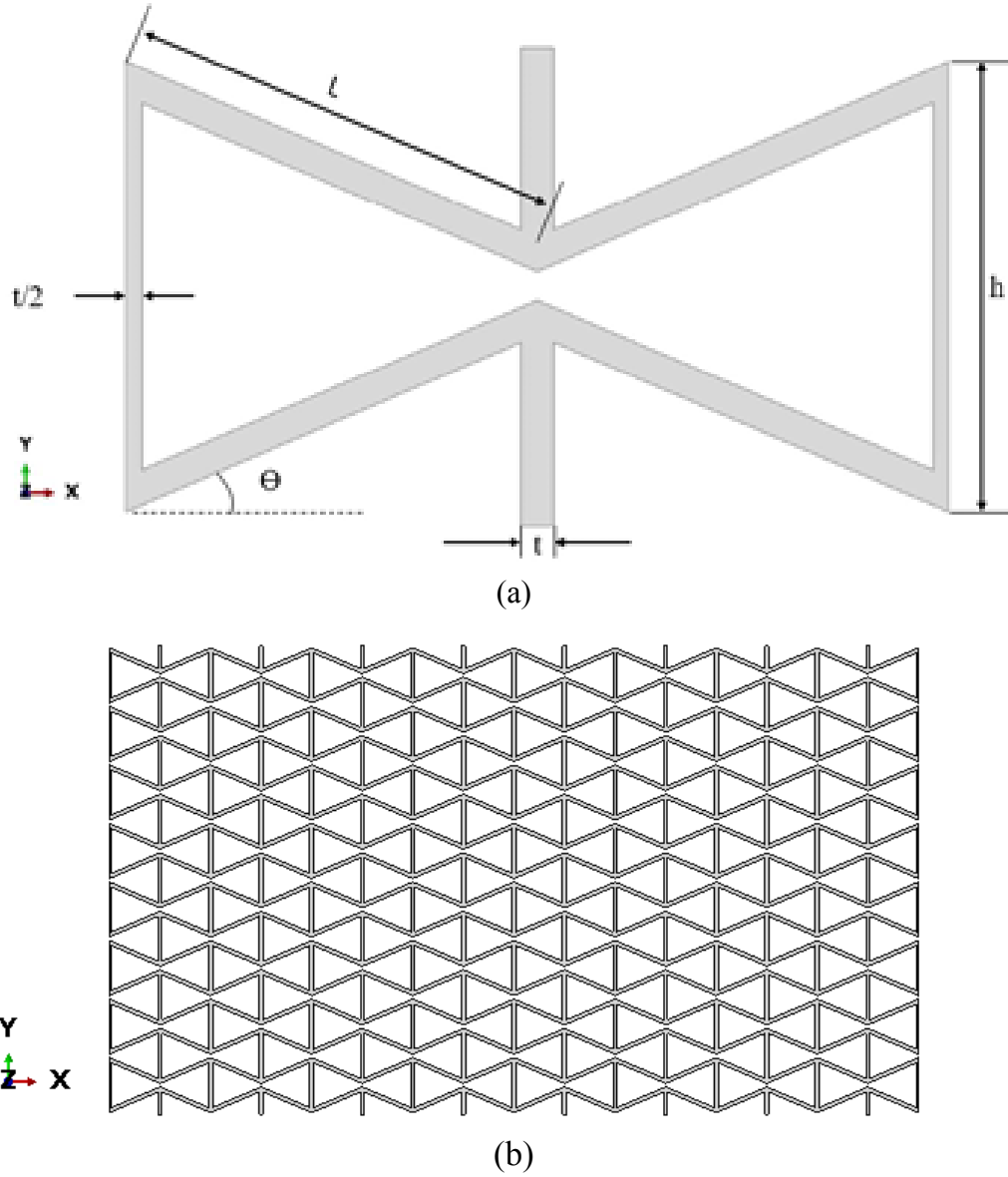


Fig. 2.1 (a) Geometric configuration of classical re-entrant cell, (b) Representative volume element (RVE) used for analysis

2.3 Research Gap

Traditional analytical or optimization approaches rely heavily on manual parameter tuning or limited geometry exploration. Modern data-driven approaches, particularly deep learning, provide an opportunity to learn the underlying mapping between geometry and mechanical response, enabling efficient inverse design.

2.4 Objectives

The objectives of this study are:

1. To develop a finite element based data acquisition pipeline for reinforced re-entrant honeycomb lattices
2. To train a deep learning model that predicts macroscopic Young's modulus and Poisson's ratio from reinforcement images
3. To implement a gradient-based optimization strategy that identifies the optimal reinforcement configuration maximizing stiffness and minimizing Poisson's ratio

2.5 Scope

This work focuses on 2D plane stress analysis of an 8x8 re-entrant honeycomb structure as the representative volume element (RVE) under small deformation. Reinforcement is modeled as inclusion of disc shaped fibres of identical material properties, merged into the base lattice. The base REH lattice is kept constant and the reinforcement is changed in the experiments.

3. Literature Review

3.1 Auxetic Material and Re-Entrant Honeycomb Structures

Auxetic materials are defined as solids that possess a negative Poisson's ratio. When auxetic materials are stretched in one direction, they expand in the direction transverse to the loading direction. It follows that if the load is reversed from stretching to compression, the materials contract in the transverse direction. Some examples of highly idealized microstructural geometries that give rise to 2D auxetic behavior are displayed in Fig. 3.1. In these examples, as well as in other geometries, some form of rotation takes place for auxetic behavior to manifest [1].

Among various auxetic topologies, the re-entrant honeycomb (REH) design remains one of the most studied due to its geometric simplicity and ability to achieve strong auxetic responses through controlled geometry parameters such as the re-entrant angle, wall thickness, and aspect ratio. The classical re-entrant geometry, first proposed by Gibson and Ashby, achieves negative Poisson's ratio through inward-pointing cell walls that rotate and stretch under load. Subsequent studies have demonstrated that geometric modifications can tailor the degree of auxeticity and stiffness, thereby enabling tunable mechanical responses [2].

3.2 Mechanical Reinforcement and Optimization of Auxetic Lattices

While auxetic behavior is beneficial, most auxetic geometries, particularly honeycomb-based ones, suffer from low in-plane stiffness, which limits their structural applicability. Researchers have attempted various reinforcement strategies to improve stiffness without losing auxeticity. These include introducing thicker ligaments, additional cross-links, or material inclusions at stress concentration zones [3].

Finite element studies have been employed extensively to evaluate how different reinforcement patterns or material gradations affect overall performance. For example, Acharya et al. [3] investigated the influence of infill based reinforcement on the mechanical response of re-entrant lattices and observed that strategic placement of stiffer regions can significantly increase the effective modulus. Similarly, Acharya et al. [4] applied novel nesting strategies to improve the stiffness.

However, these traditional approaches rely on iterative finite element runs and manual parameter tuning, which become computationally expensive as the design space grows.

Moreover, the nonlinear coupling between geometry, material distribution, and global auxetic behavior makes gradient-based optimization difficult to perform directly in the FE domain.

3.3 Machine Learning for Structure-Property Prediction

The rise of machine learning (ML) and deep neural networks (DNNs) has enabled data-driven approaches to predict and design materials with desired properties. In particular, convolutional neural networks (CNNs) have shown strong capability in learning spatial correlations from image-based representations of microstructures.

In the context of mechanical metamaterials, Chang et al. [5] used CNNs to predict the elastic constants of porous materials directly from binary images of their microstructure. Gu et al. [6] further demonstrated that deep generative models can be trained to perform inverse design, generating new microstructures targeting specific stiffness and strength values. Farhad et al. [7] integrated ML models with FE data to develop a fast surrogate framework for composite design, reducing computational cost while maintaining physical accuracy.

Despite these advances, there remains a gap in applying deep learning to auxetic reinforcement optimization. Most works either focus on topology optimization of non-auxetic lattices or on isotropic material microstructures. The integration of FE-simulated datasets with CNN-based prediction for auxetic honeycombs remains relatively unexplored.

3.4 Reinforcement Representation and Data-driven Optimization

Image-based reinforcement representation, where the geometry or reinforcement topology is encoded as a grayscale or binary image, offers a natural interface between physical simulations and CNN architectures. The spatial layout of reinforcements can be effectively captured in this manner, allowing the CNN to learn physically meaningful features.

Furthermore, differentiable surrogate models, once trained, can be integrated into optimization loops where gradient-based algorithms are applied to the input representation (image or geometry parameters) to search for configurations that optimize a chosen cost function. This provides a bridge between traditional FE simulations and modern ML-based inverse design frameworks.

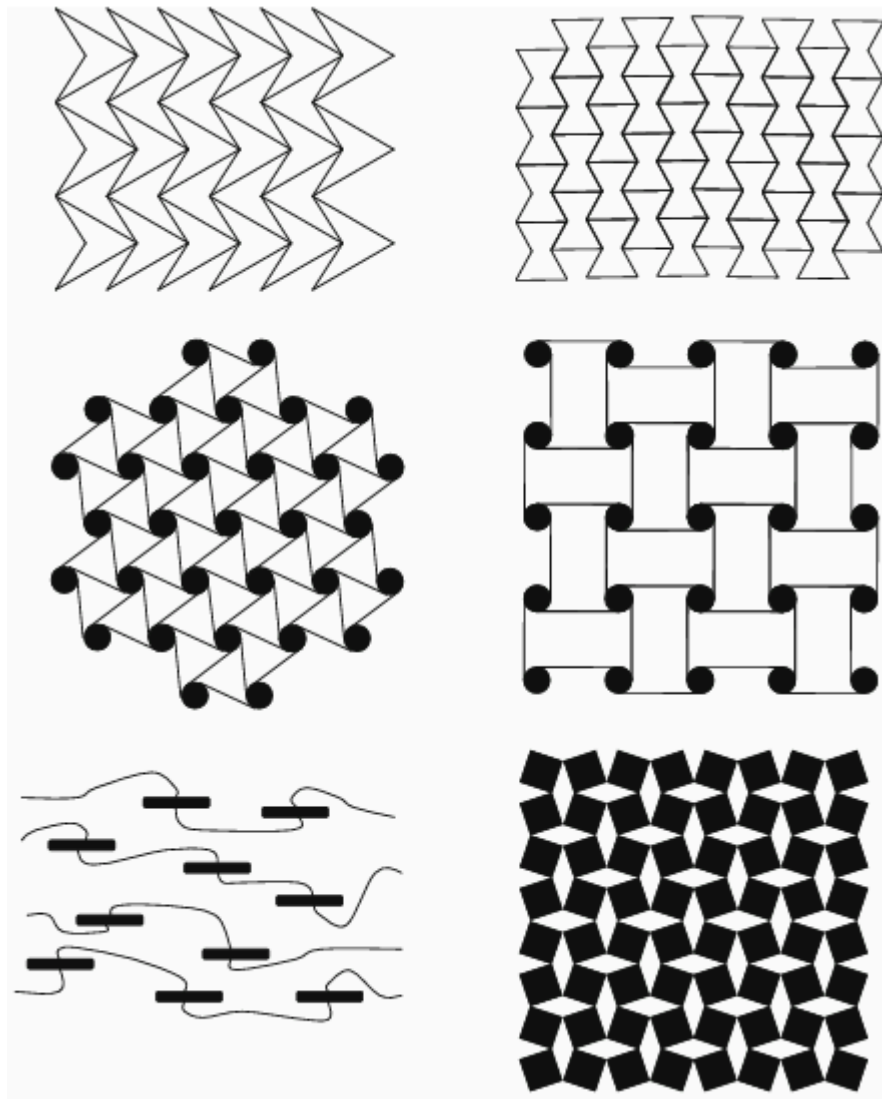


Fig. 3.1 A few highly simplified geometries that give auxetic behavior [\[1\]](#)

4. Methodology

4.1 Overview

This research employs a two-stage approach:

1. Generation of FE simulation data for various reinforcement configurations
2. Development of a CNN model for property prediction and subsequent optimization via gradient descent

4.2 Finite Element Model Setup

An 8×8 representative volume element (RVE) of classical re-entrant honeycomb (REH) cells is modeled in Abaqus/CAE 2025. Each cell is defined by:

$$\theta = 23^\circ, t = 3 \text{ mm}, l = h = 42 \text{ mm}$$

Material: Steel

$$E_s = 210 \text{ GPa}, \nu_s = 0.3$$

Analysis type: Plane Stress

Element type: CPS8R (8-node biquadratic, reduced integration)

Mesh size: $t / 4 = 0.75 \text{ mm}$

A zoomed in view of the mesh generated for classical re-entrant lattice under the above conditions is depicted in Fig. 4.1.

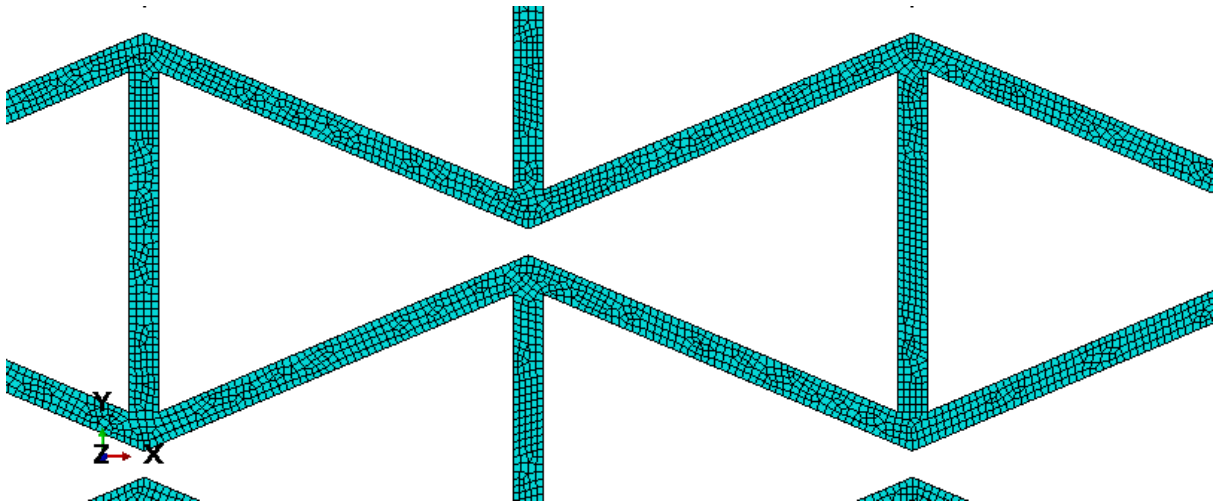
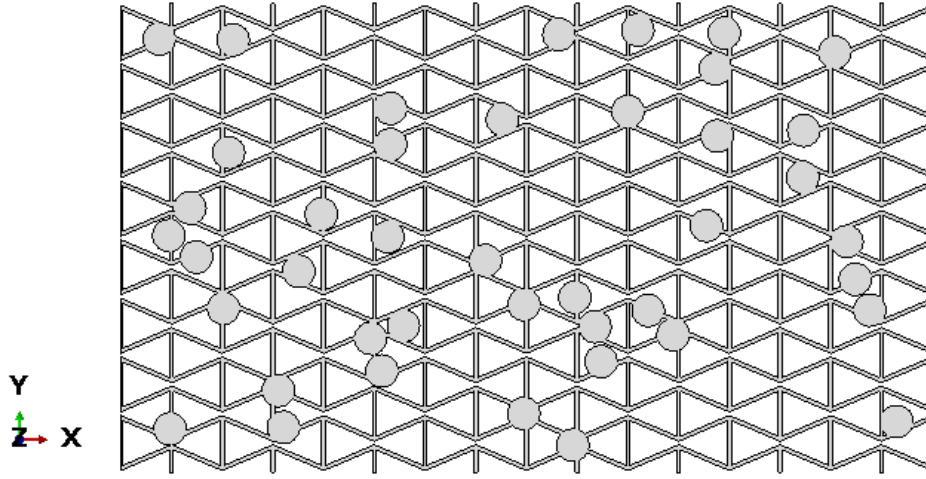


Fig. 4.1 Zoomed in view of the generated mesh with CPS8R meshing element

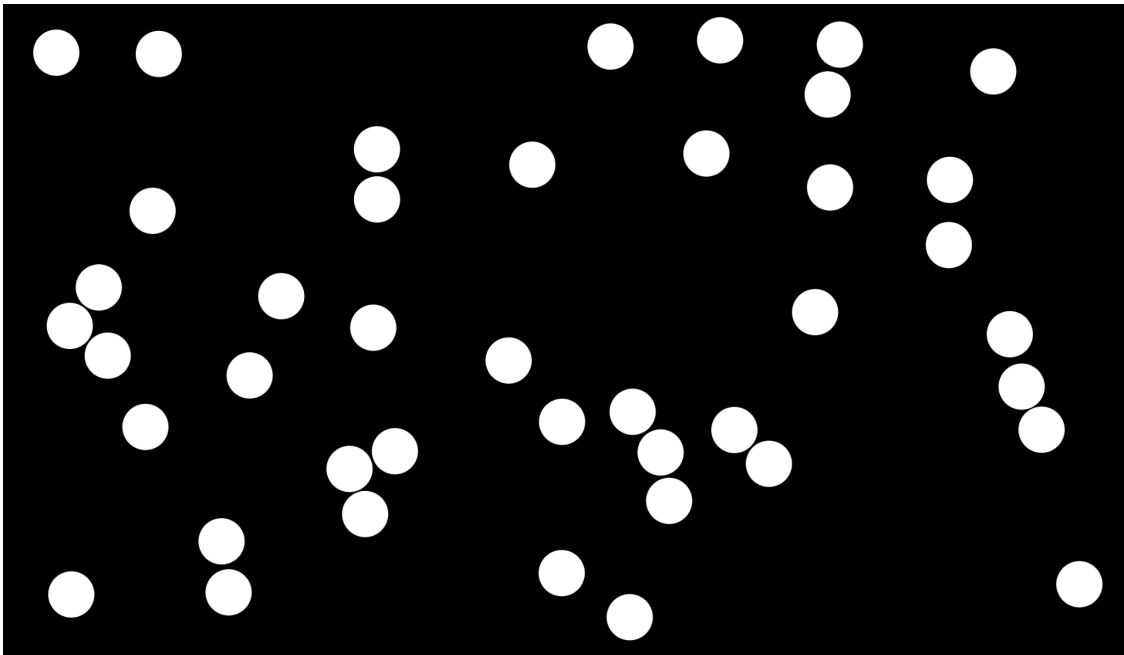
4.3 Reinforcement Modeling

Circular disc reinforcements are randomly placed within the lattice ensuring no overlap [8]. To ensure that the discs are always attached to the structure and won't float in the vacuum, the

disc sizes were restricted to be larger than the in-radius of the triangular cavity of the re-entrant cell. The number of reinforcements n and their radius r are varied across simulations. The discs are merged with the lattice to ensure structural continuity. Each configuration is represented by a binary image I (white pixels: reinforced zones). For instance the reinforced structure shown in Fig. 4.2(a) gets encoded into a black and white image with regions of reinforcement brightened as shown in Fig. 4.2(b).



(a)



(b)

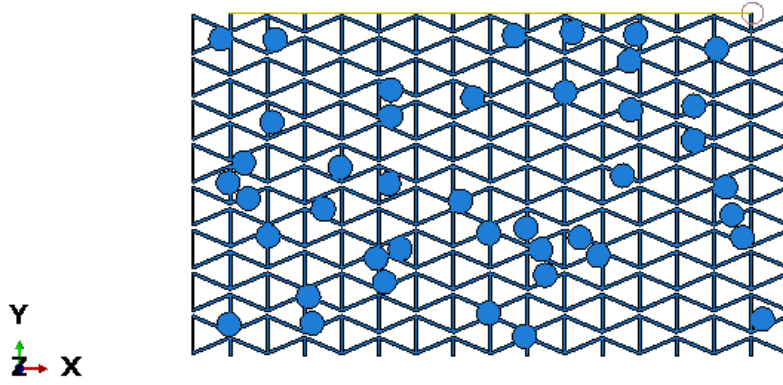
Fig. 4.2 (a) Reinforced structure for parameters $n = 42$, $r = 12.521$, (b) Corresponding image representation for the reinforcement of the re-entrant lattice

4.4 Boundary Conditions and Loading

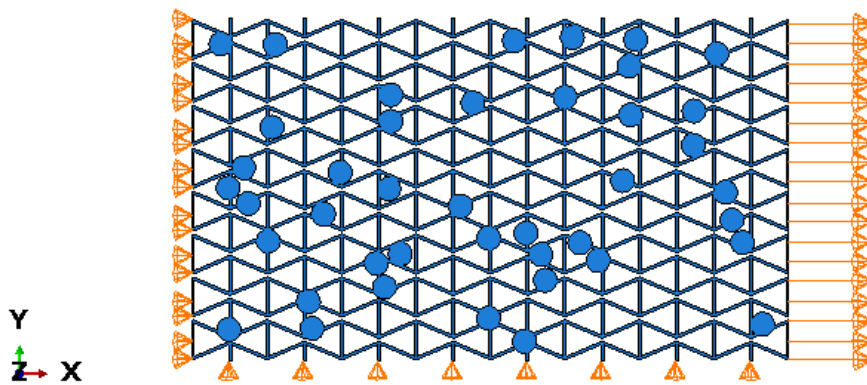
Periodic Boundary conditions were employed as follows [3]:

- Left face: $U_x = 0$
- Bottom face: $U_y = 0$
- Top face: Uniform U_y via kinematic coupling
- Right face: 1% tensile strain ($\epsilon_{xx} = 0.01$)

This ensures quasi-periodic deformation representative of bulk material response. The boundary conditions and coupling constraints on the mesh described above are depicted in Fig. 4.3(a)-(b)



(a)



(b)

Fig. 4.3 (a) Kinematic coupling of upper edge, (b) Boundary conditions and application of 1% strain along x-direction

4.5 Property Extraction

From the FE results, the displacement field U and reaction force field RF are utilized to extract the macroscopic stress and macroscopic strain values as written in Eq. 4.1 which in turn are utilized to extract Young's modulus E and poisson's ratio ν as written in Eq. 4.2.

$$\varepsilon_{yy} = \frac{\overline{U_{y,top}}}{H}, \quad \sigma_{xx} = - \frac{\Sigma RF_{x,left}}{Ht_s} \quad (4.1)$$

$$\nu = - \frac{\varepsilon_{yy}}{\varepsilon_{xx}}, \quad E = \frac{\sigma_{xx}}{\varepsilon_{xx}} \quad (4.2)$$

where H is total height, t_s is section thickness.

The deformed state of the mesh is shown in Fig. 4.4(a) and an image of it overlaid on top of undeformed mesh is shown in Fig. 4.4(b) which clearly shows mesh expansion in both x and y directions as expected by an auxetic mesh.

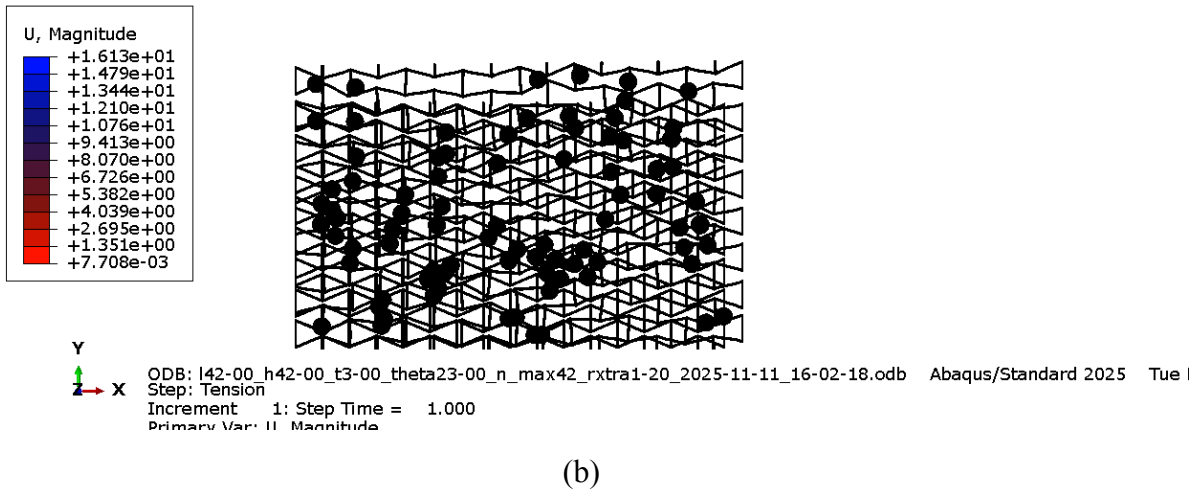
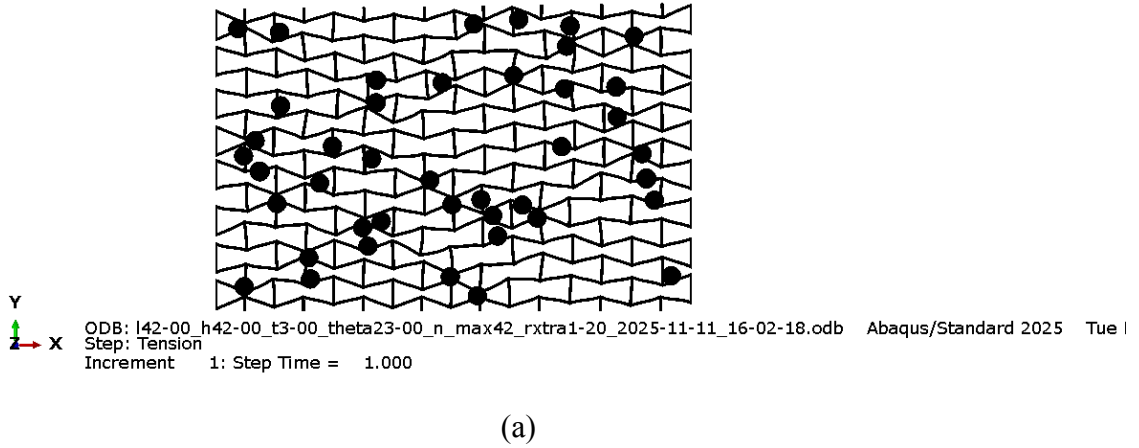
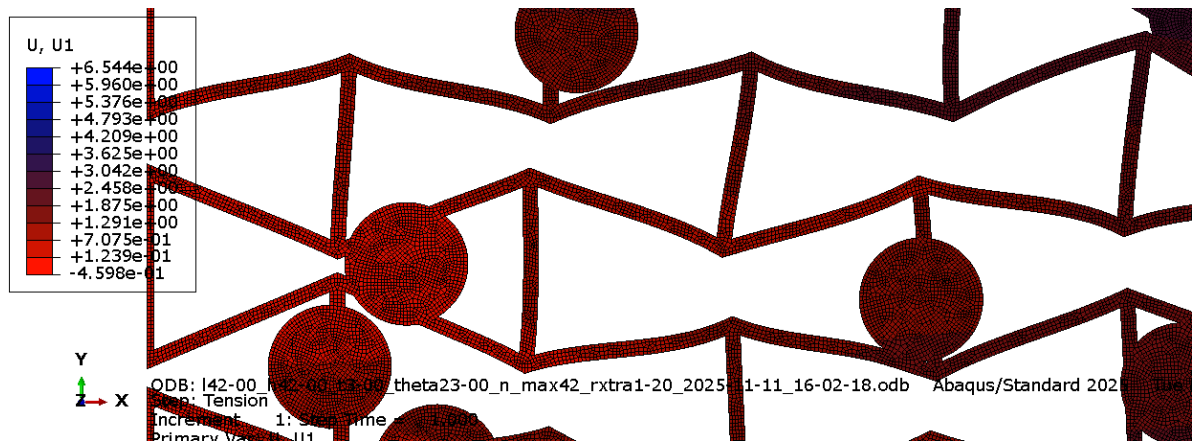


Fig. 4.4 (a) deformed state of the mesh (b) overlaid deformed and undeformed mesh

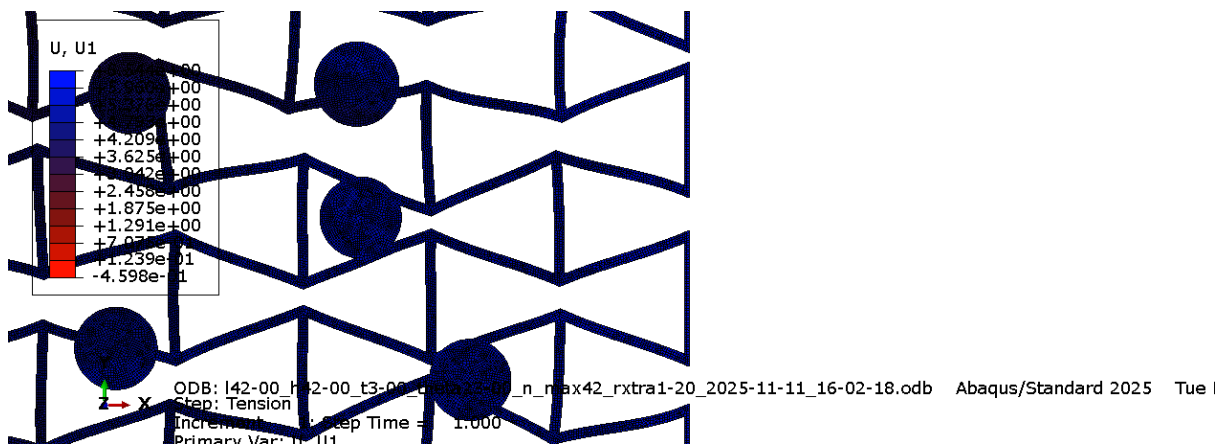
Below are some figures that show zoomed in view of the mesh along the edges. Firstly, in Fig. 4.5, a colormap of reaction force along x direction is shown along the left face. This is almost uniform on the whole of the mesh body. Fig. 4.6(a)-(c) depicts the colormap of displacement in x direction of mesh elements. The left edge (a) is immobile in x-direction as expected by the displacement boundary condition. All elements on the right edge (b) are displaced towards the right by an amount 1% of the strain consistent with the input strain. The top edge (c) has a gradient going from red to violet (immobile to free). Further the Fig. 4.7(a)-(c) depicts the colormap of displacement in y direction of mesh elements whereby as expected the bottom edge (b) is immobile throughout and the top edge (a) has some positive displacement uniform throughout the edge as expected by the kinematic coupling constraint on the top edge.



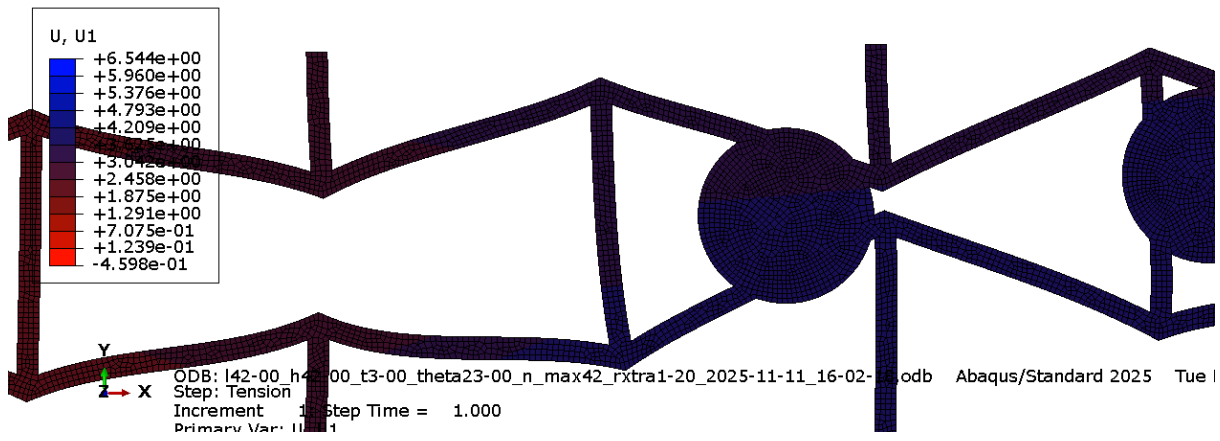
Fig. 4.5 RF1 plot for a part of the left side



(a)

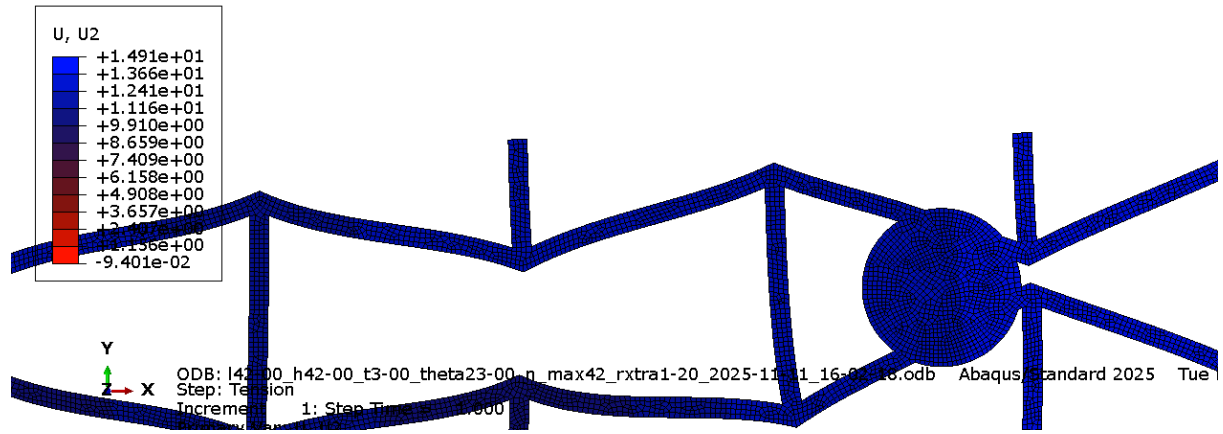


(b)

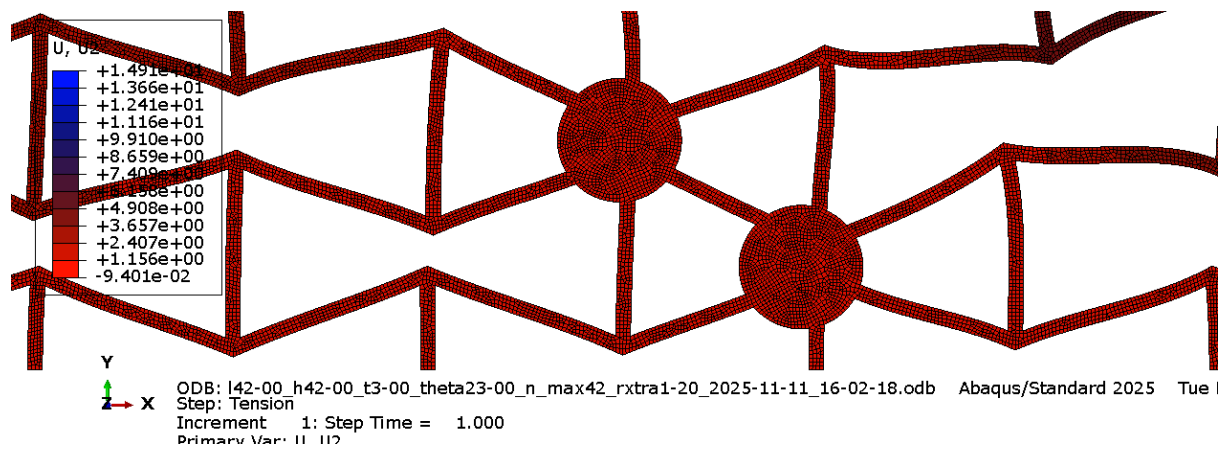


(c)

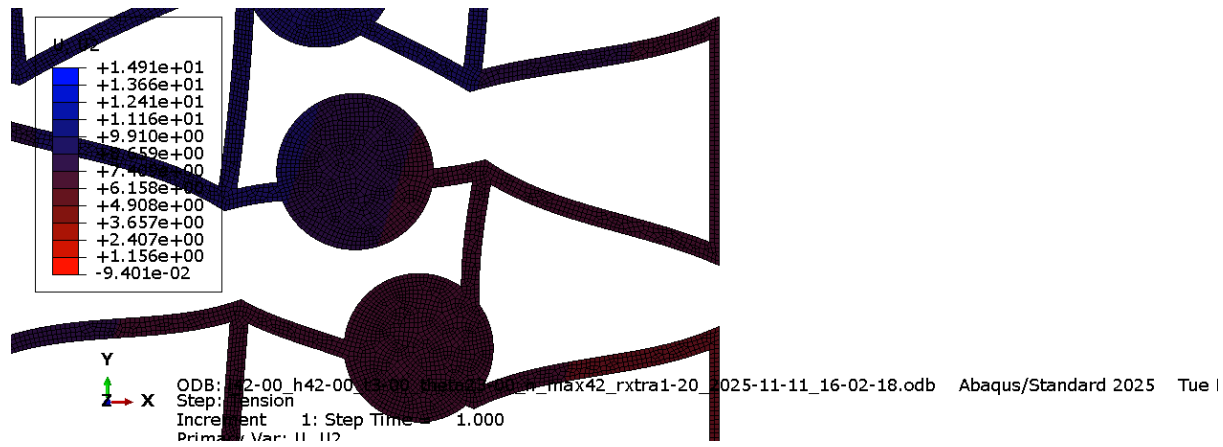
Fig. 4.6 (a) U1 for left face, (b) U1 for right face, and (c) U1 for top face



(a)



(b)



(c)

Fig. 4.7 (a) U2 for top face, (b) U2 for bottom face, and (c) U2 for right face

4.6 Data Acquisition Pipeline

A Python-Abaqus script automates random reinforcement generation (image I , number of discs n , radius of each disc r), FE simulation is on the generated structure and fetching of the

field outputs, property computation and data storage. The dataset format is $(I, n, r) \rightarrow (E, v)$ mappings.

Each entry corresponds to a distinct reinforcement configuration and its macroscopic response. In the context of modelling, we collect E/E_s values as the labels for each input as they create independence over the material chosen while performing the simulation. For any linear homogeneous isotropic material the values of E/E_s would be expected to be similar.

4.7 Model Architecture

The model consists of

1. Convolutional Encoder: This encoder uses sequential convolutional blocks with batch normalization after each convolution operation. We use 2 small architectures shown in Fig. 4.8(a)-(b) to limit the computational cost and make it realistic with respect to dataset size. This component maps the reinforcement images to a latent vectorial representation using the textural local patterns.

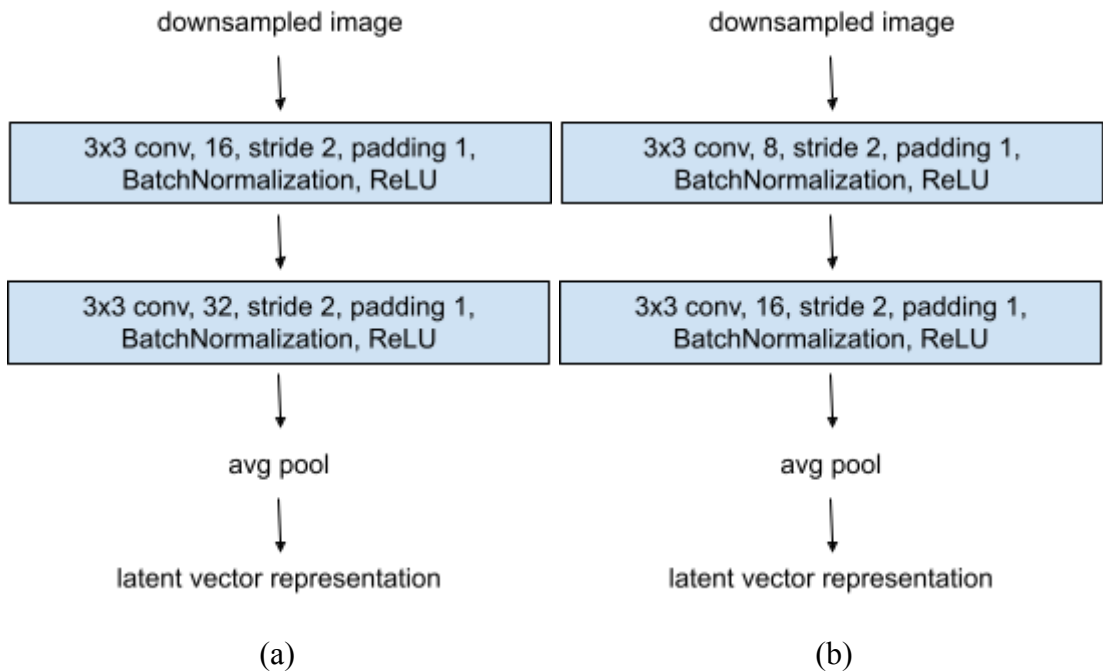


Fig. 4.8 Convolutional Image Encoder for Architecture (a) 1, (b) 2

Each reinforcement image that was collected was downsampled first using box downsampling to decrease the input size before feeding it into the encoder so as to massively reduce the computations for limited compute.

2. MLP Head: This is a sequence of multiple fully connected layers (multi-layer perceptron) that maps the latent vector representation of input reinforcement to the two values we try to model, i.e., E/E_s and v . For the two convolutional encoders, we chose corresponding MLP heads as shown in Fig. 4.9(a)-(b).

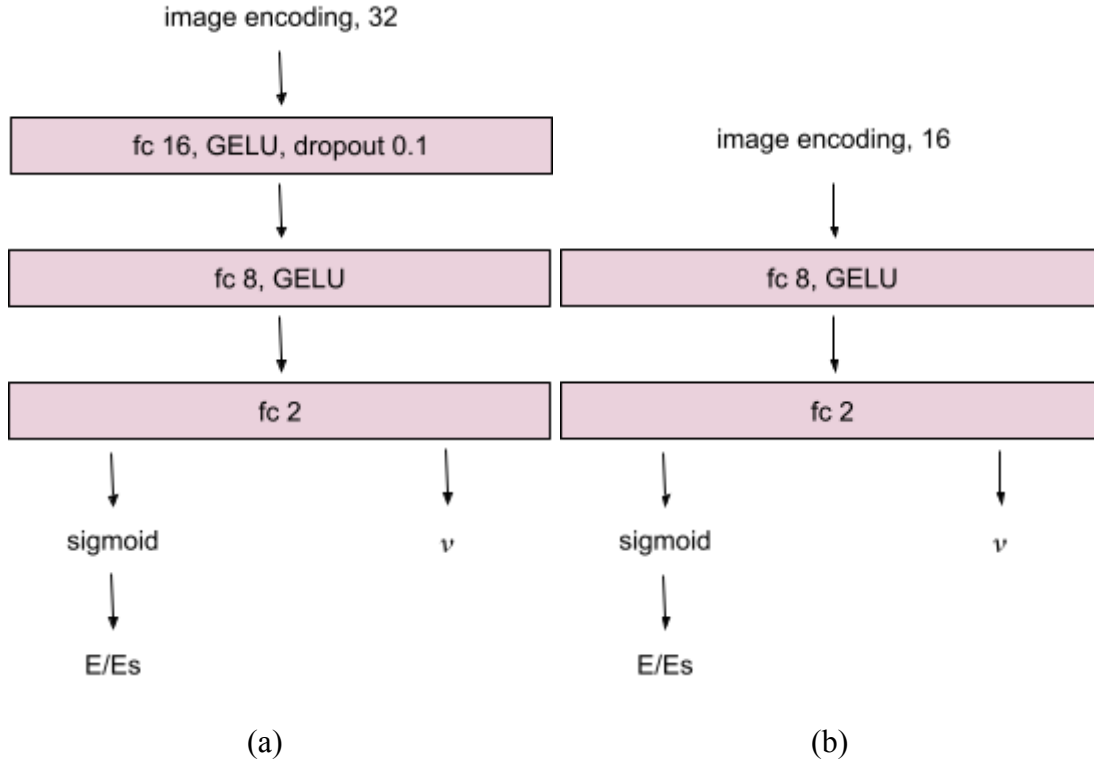


Fig. 4.9 MLP head for Architecture (a) 1, (b) 2

We will refer to the two architectures as Architecture 1 and Architecture 2. The number of training parameters are 5578 in Architecture 1 and 1450 in Architecture 2.

4.8 Training Procedure

The data available would be split into train and test sets. The training would be performed on the train set and the model would be tested on the test set. The split performed in our case is such that for each (n, r) configuration, one of the data points is randomly selected to be a part of the test set while the rest belong to the train set.

Training tools would be as follows:-

- Loss function: Smooth L1 Loss between predicted and true E and v

Smooth L1 Loss is a hybrid loss function that acts like L2 loss when the absolute error is small (below a threshold β) and L1 loss when the error is large, offering a balance

between outlier robustness and smooth convergence. A standard Smooth L1 Loss function can be defined as in the Eq. 4.3.

$$L(x) = \begin{cases} 0.5(x - y)^2, & \text{if } |x - y| \leq 0.5 \\ |x - y| - 0.5, & \text{otherwise} \end{cases} \quad (4.3)$$

- **Optimizer:** Adam which combines the benefits of AdaGrad and RMSProp. It computes individual, adaptive learning rates for different parameters by estimating the first and second moments of gradients making convergence smoother.
- **Dataset split:** The split strategy mentioned before splits the dataset in nearly 80:20 train-test split.

The model is trained until the accuracy is reasonably good and no further improvements are observed in subsequent epochs, i.e., the model hits saturation over the given training data. It would be assumed that the deep learning model is fairly close to the actual mapping function that takes the reinforcement image as input.

The trained model's performance is assessed using Mean Absolute Error (MAE) Eq. 4.4 and R-squared error Eq. 4.5:

$$MAE = \frac{1}{N} \sum |y_i - \hat{y}_i| \quad (4.4)$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (4.5)$$

Here y_i is observed value, $\hat{y}_i$ is predicted value and $\bar{y}$ is the mean of observed values.

The regularization of the model is important to ensure that the model doesn't learn some undesired patterns from the data which is ensured using batch normalization and dropout layers in model architecture.

4.9 Optimization Framework

After training, the model acts as a differentiable surrogate for FE simulations. The goal is to find the reinforcement configuration that maximizes stiffness (E) and minimizes Poisson's ratio (ν). A simple cost function that can capture the tradeoff can be defined as in Eq. 4.6.

$$J(I, n, r) = \omega(-\alpha E/E_s + \beta^{v+1} + \gamma v) \quad (4.6)$$

where $\alpha, \beta, \gamma, \omega$ are tunable parameters balancing stiffness and auxeticity.

The choice of exponential w.r.t. v ensures that its heavily penalized when v is nearly 0 which is near loss of auxeticity of the lattice. When in the domain of negative v , its still considerable but its effects on the gradients should be significantly low as the lattice would still be auxetic and in such domain of negative v , the focus should be shifted heavily on maximizing E/E_s .

Adjusting α and γ would enable us to set a desired tradeoff between the two in this regime. ω is just a scalar to ensure gradient values are not too low and tune the cost as a whole.

A plot of this function for a specific case is shown in Fig. 4.10.

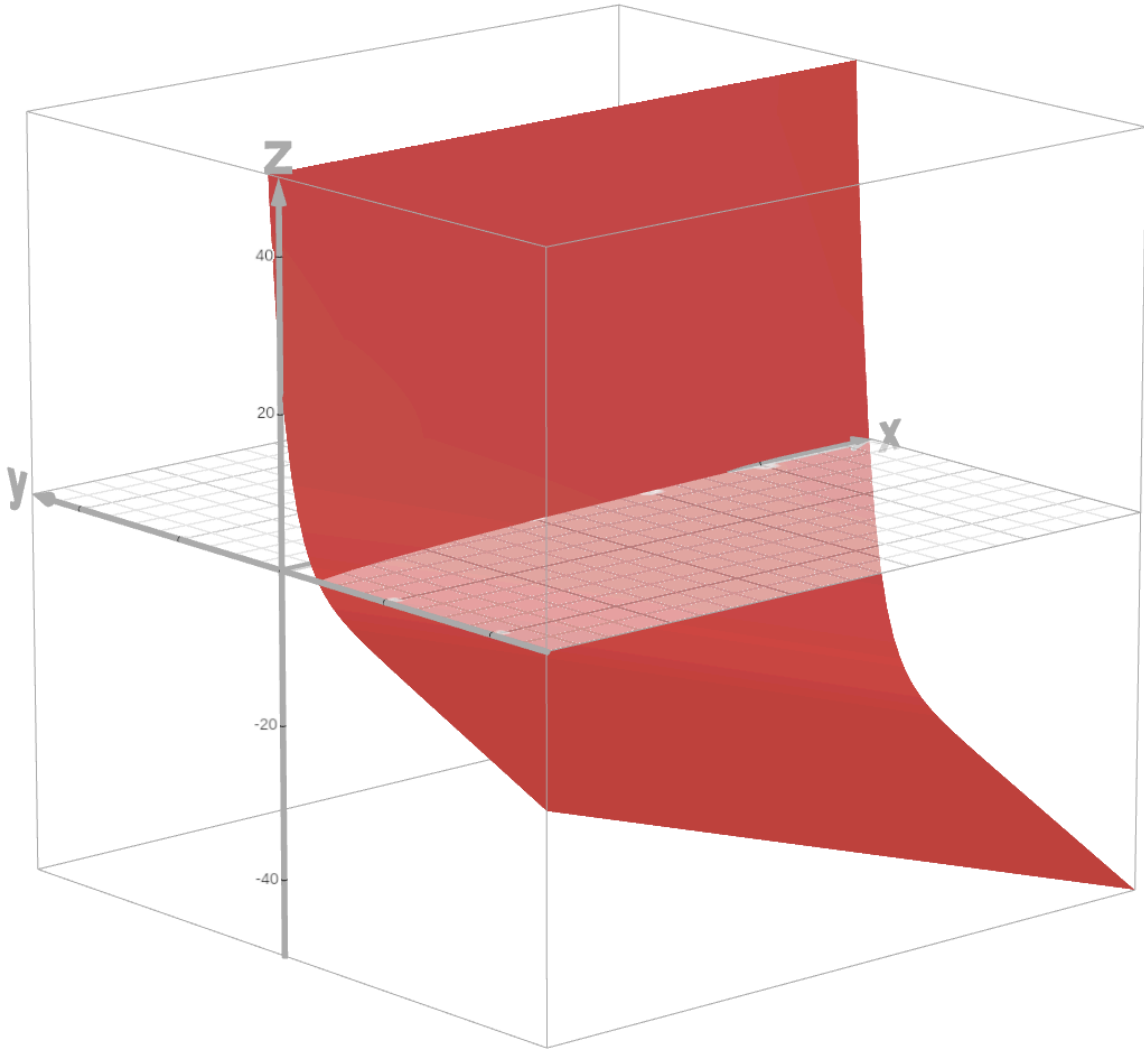


Fig. 4.10 A plot for the optimizer penalty function that captures the strength-auxeticity tradeoff with $\alpha = 30.0$, $\beta = 25.0$, $\gamma = 4.0$, $\omega = 1.0$

4.10 Gradient-Based Search

Starting from an initial reinforcement image I , gradient descent is performed on the cost function crafted in Eq. 4.6 as shown in Eq. 4.7.

$$I \leftarrow \sigma(I - \eta \text{grad}(J)) \quad (4.7)$$

where η is the rate of descent.

Here, the trained CNN provides differentiable gradients w.r.t. each input pixel thus enabling a pixel level optimization and generating the optimal reinforcement according to the patterns learned by the model.

Multiple starting points can be chosen and a set of minima points can be obtained local to those starting points. This will be done by performing this experiment several times with random starting images.

For the process, we first make an image with random pixel values and then perform the descent step. We then apply a sigmoid function on it and scale it by 255, which is the highest value any pixel can take, ensuring the image remains in valid range. Basic clipping is not used to ensure that gradients don't die out due to clipping.

4.11 Optimal Configuration

The optimized reinforcement image obtained from gradient-based optimization exhibited high-frequency noise and scattered pixel activations. These artifacts arise due to pixel-level optimization in a high-dimensional space and do not directly correspond to physically meaningful structures. To extract coherent reinforcement regions, a density-based segmentation approach was employed.

- **Density Estimation:** The binary reinforcement image is first converted into a continuous density field using Gaussian smoothing as depicted in Eq. 4.8.

$$D(x, y) = (G_\sigma * I)(x, y) \quad (4.8)$$

where G_σ is a Gaussian kernel with standard deviation σ and the operation $*$ denotes convolution.

This operation aggregates local pixel information and transforms sparse clusters of white pixels into continuous high-density regions.

- **Threshold-Based Segmentation:** The density field is converted back to a binary image using thresholding using Eq. 4.9.

$$S(x, y) = \begin{cases} 1, & \text{if } D(x, y) > \tau \\ 0, & \text{otherwise} \end{cases} \quad (4.9)$$

where τ is a threshold parameter

This step separates regions with high reinforcement density from background regions, effectively eliminating isolated noise.

- Morphological Refinement: To further improve spatial coherence, morphological closing is applied as depicted in Eq. 4.10.

$$S'(x, y) = (S(x, y) \oplus B) \ominus B \quad (4.10)$$

where $\oplus$ denotes dilation, $\ominus$ denotes erosion and B is a structuring element (kernel)

This operation fills small gaps within regions and smoothens boundaries without significantly altering global structure.

This approach converts noisy pixel-level outputs into structured reinforcement regions suitable for interpretation and further analysis.

Morphological refinement also allows us to define the kernel size such that each independent region is bigger than the critical disc ensuring no structural collapse. The kernel size is set such that its corresponding size in the lattice domain is bigger than the in-radius of the inner triangular rib region of the REH unit cell.

5. Results and Discussion

5.1 Curation of Dataset

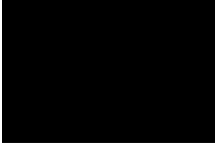
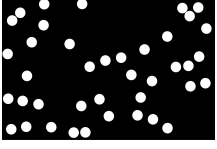
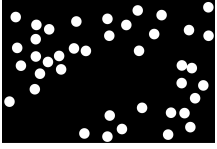
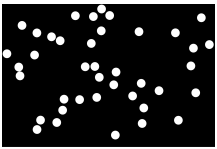
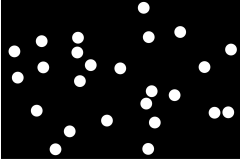
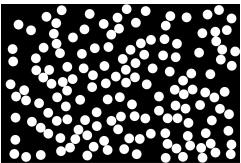
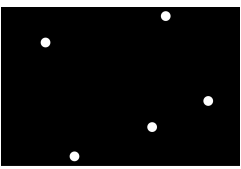
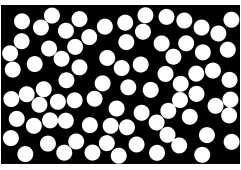
The base re-entrant lattice exhibits significantly lower elastic modulus compared to the section material and has a negative Poisson's ratio as expected. Also, when it is reinforced, the elastic modulus increases and poisson's ratio increases (becomes less negative) with increase in the amount of reinforcement.

Table 5.1 shows the results obtained for classical re-entrant honeycomb lattice 8x8 RVE and the same reinforced with few reinforcement parameters against the input reinforcement (stored as an image), number of disc fibers and their radii.

For the data collected for our experimentation, we varied n from 5 to 150 in steps of 5 except the base experiment of no reinforcement. The amount by which the radius of each disc is more than the critical in-radius value discussed before is varied from 0.0 to 0.8 in steps of 0.2. Each (n, r) configuration is run 8 times to get 8 different random placements of discs for the same n and r . In total, the dataset size thus becomes $40 * 30 * 8 + 1 = 9601$ data points.

These results were plotted on 3D axes as shown in Fig. 5.1. For each (n, r) pair, variation in properties due to changes in reinforcement is significant and the deep learning model to be developed has a lot to learn from these images. Also, the surface plot of the values of these mechanical properties averaged over all 8 experiment for each (n, r) shows a rise in both properties as any of n or r is increased.

Table 5.1 Few $(I, n, r) \rightarrow (E/E_s, \nu)$ observed from the collected dataset

Reinforcement Image	Number of disc fibers	Radius of each disc fiber (mm)	Elastic modulus (MPa)	Poisson's ratio	E/E_s
	0	0.00	846.227	-3.4591	0.00403
	40	14.999	1509.687	-3.3577	0.007189
	40	14.999	1303.141	-3.3956	0.006205
	40	12.399	1078.031	-3.4277	0.005133
	25	14.999	1102.383	-3.4240	0.005249
	150	12.399	4142.262	-2.9972	0.019725
	5	12.399	912.326	-3.4476	0.004344
	80	20.399	18416.825	-1.1138	0.087699

This is consistent behavior, as the degree of reinforcement increases, the elastic modulus of the lattice increases but simultaneously the poisson's ratio becomes less negative, i.e., the structure becomes less auxetic. Another important observation from the plot is the variance increases a lot as the value of n increases across multiple experiments for each value of r .

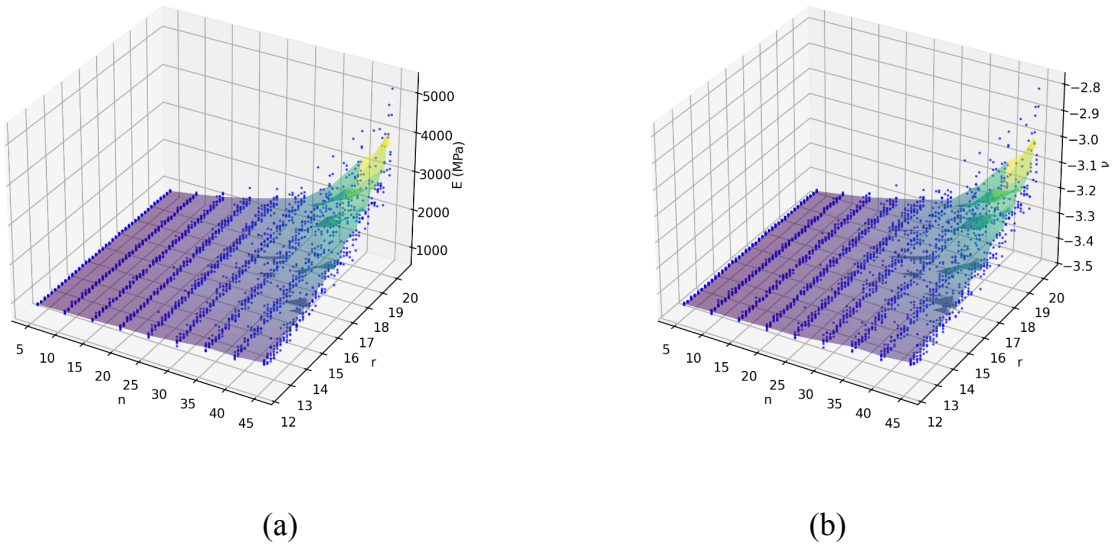


Fig. 5.1 (a) Plot of E (MPa), (b) Plot of ν for different (n, r) values of REH RVE reinforcement

The specific elastic modulus is defined as $E/\rho E_s$ where ρ is the mass density of the lattice which can be calculated as $\rho_s v$ where v represents the volume fraction of the actual material in the cut out merged lattice to the 8×8 space that it occupies. The design space for the collected dataset is plotted as a plot of specific stiffness vs poisson's ratio as shown in Fig. 5.2 where the specific density is scaled up by ρ_s which is a constant to have E/vE_s on the x-axis. Such a distribution with competing variables that guide the design decisions is called the Pareto Front.

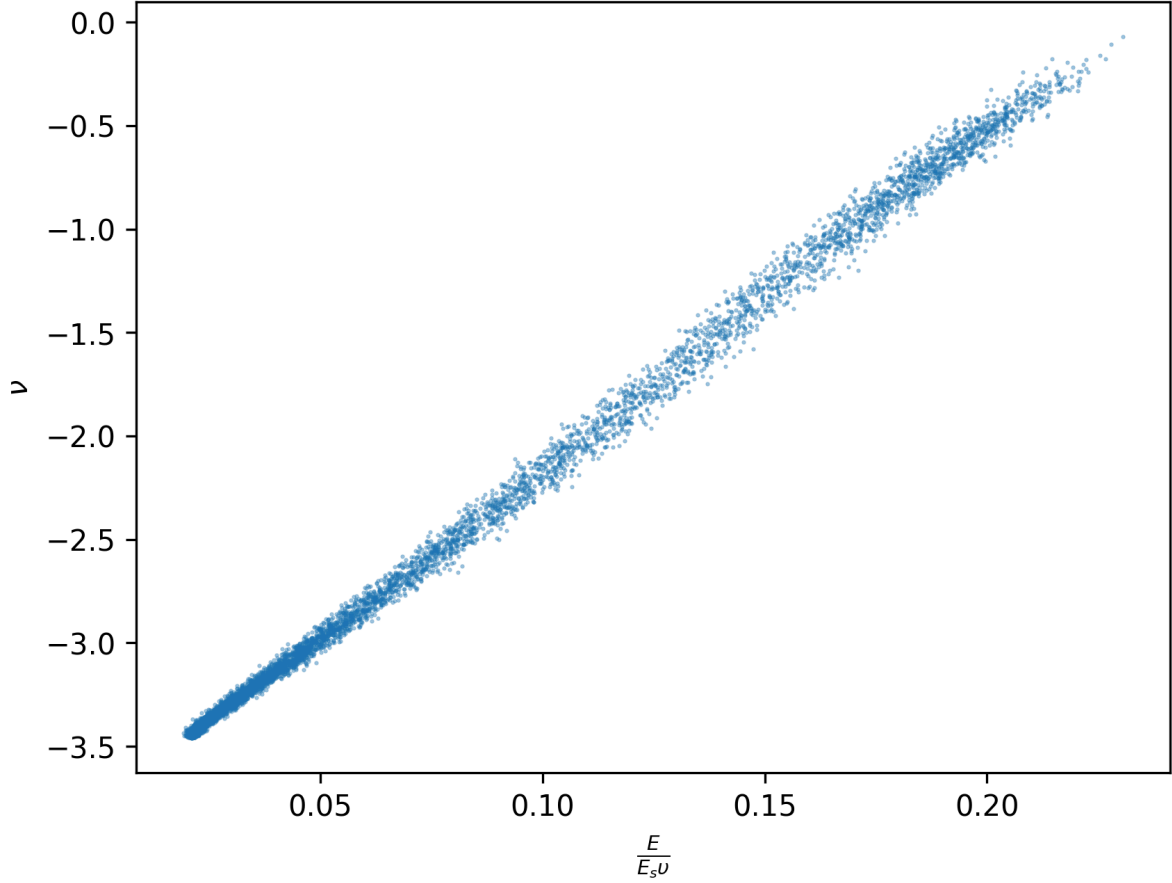


Fig. 5.2 Pareto Front and the Design Space of the curated dataset

5.2 Trained Deep Learning Model

Using the training procedure discussed in section 4.8 both architectures are trained on the image to $(E/E_s, v)$ data points. Fig 5.3 (a)-(b) shows all the data points predictions vs true value plots. The model provides decent predictions with even distribution about identity line in prediction vs true value plane.

Both models learn well from the dataset thus giving good results on the test set as well shown in orange blips in Fig 5.3 (a) and (b).

The Smooth L1 Loss values for the trained models on test set are computed to be 1.07 for architecture 2 which is quite small and 0.99 for architecture 1 which is much bigger than architecture 2. This shows the additional layers and parameters in the first architecture doesn't make much significant improvement over the predictive power of the model. The dataset saturates the predictive power of this class of model to a Smooth L1 Loss near 1. We compute

MAE and R^2 scores to further evaluate both of these models. The observed values are recorded in Table 5.2.

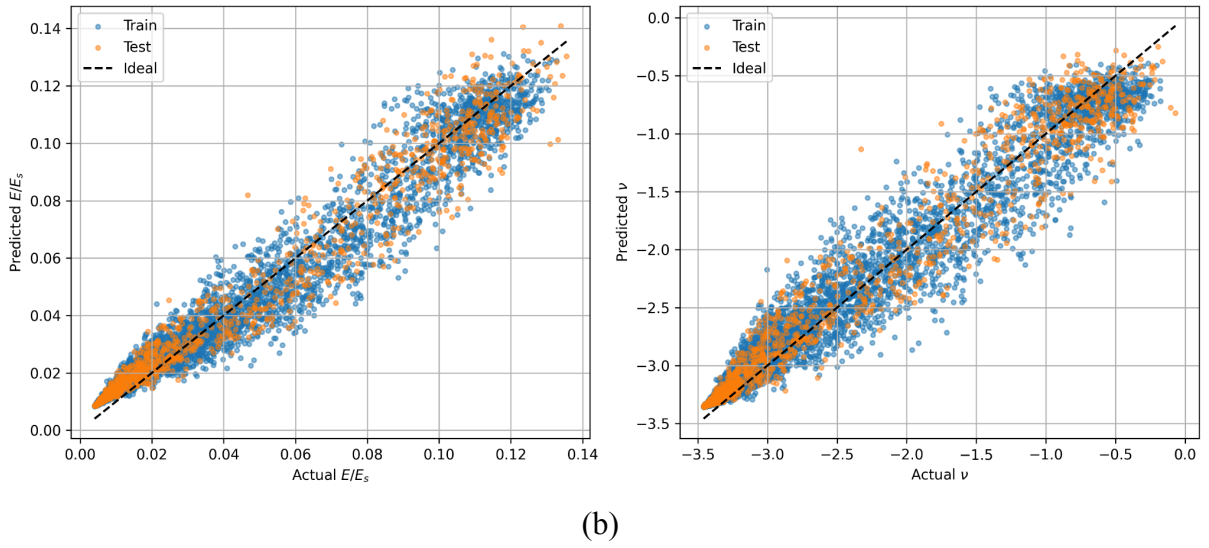
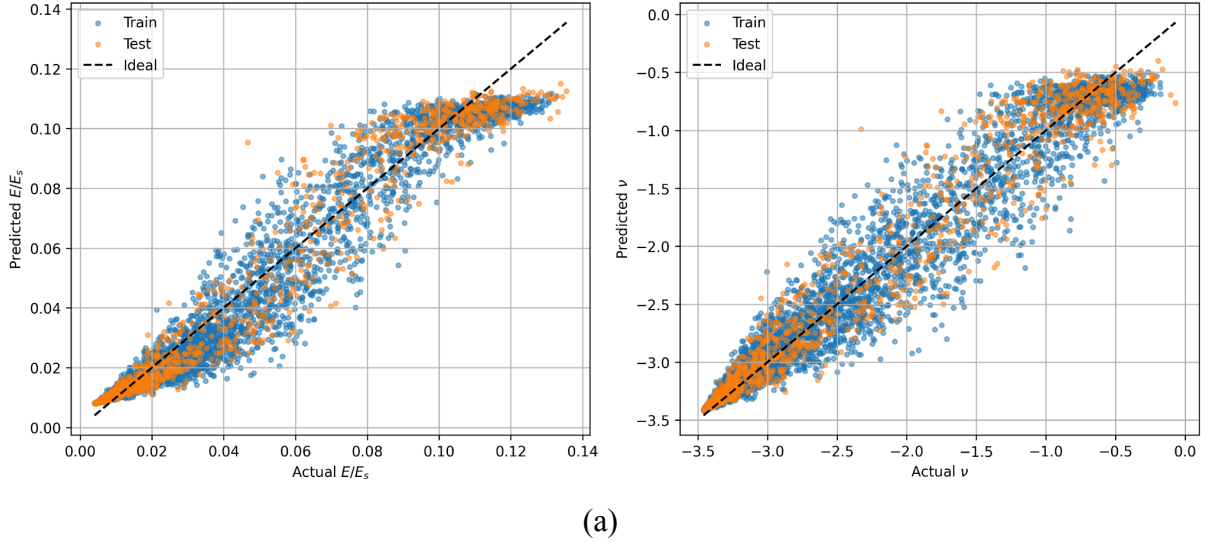


Fig 5.3 Prediction vs True value plot for Architecture (a) 1 and (b) 2 with first plot for E/E_s and second one for ν

The values of MAE and R^2 scores suggest a remarkable fit on both test and train set for both the deep learning architecture, i.e., for both shallow and small and bigger and deeper nets. Both of the models fit really well on a similar extent.

Table 5.2 MAE and R2 scores for the trained CNN architectures

Architecture	Metric type	Set evaluated on	Mechanical property	Value
1	MAE	Train	E/E_s	0.0050
1	MAE	Train	ν	0.1249
1	MAE	Test	E/E_s	0.0057
1	MAE	Test	ν	0.1498
1	R ²	Train	E/E_s	0.9601
1	R ²	Train	ν	0.9597
1	R ²	Test	E/E_s	0.9600
1	R ²	Test	ν	0.9586
2	MAE	Train	E/E_s	0.0057
2	MAE	Train	ν	0.1500
2	MAE	Test	E/E_s	0.0063
2	MAE	Test	ν	0.1692
2	R ²	Train	E/E_s	0.9586
2	R ²	Train	ν	0.9548
2	R ²	Test	E/E_s	0.9598
2	R ²	Test	ν	0.9555

5.3 Gradient Based Search on Trained Surrogate

For trained models of both the architectures, the optimization loop was run on the designed optimization penalty. Here we show the optimization on the specific case with $\alpha = 30.0$, $\beta = 25.0$, $\gamma = 4.0$, $\omega = 100.0$. For multiple starting reinforcement images, the descent loop is run and the outputs are observed.

We show results of 4 different experiments run on multiple randomly generated images. The optimization loop is run for 100 epochs for all of these experiments.

Experiment 1: Fig. 5.4 (a) shows the initial image and Fig. 5.4 (b) shows the optimized image from architecture 1 and Fig. 5.4 (c) shows the optimized image from architecture 2.

For architecture 1, the initial image had a penalty score of 7968.0918, the optimized image gave penalty score of -1092.5857 with $E/E_s = 0.0243$ and $v = -2.5509$.

For architecture 2, the initial image had a penalty score of 153200.4688, the optimized image gave penalty score of -1061.6783 with $E/E_s = 0.0045$ and $v = -2.6221$.

Experiment 2: Fig. 5.5 (a) shows the initial image and Fig. 5.5 (b) shows the optimized image from architecture 1 and Fig. 5.5 (c) shows the optimized image from architecture 2.

For architecture 1, the initial image had a penalty score of 7900.0410, the optimized image gave penalty score of -1099.5858 with $E/E_s = 0.0236$ and $v = -2.5738$.

For architecture 2, the initial image had a penalty score of 150719.8750, the optimized image gave penalty score of -1064.2727 with $E/E_s = 0.0044$ and $v = -2.6287$.

Experiment 3: Fig. 5.6 (a) shows the initial image and Fig. 5.6 (b) shows the optimized image from architecture 1 and Fig. 5.6 (c) shows the optimized image from architecture 2.

For architecture 1, the initial image had a penalty score of 8464.8457, the optimized image gave penalty score of -1095.1536 with $E/E_s = 0.0240$ and $v = -2.5597$.

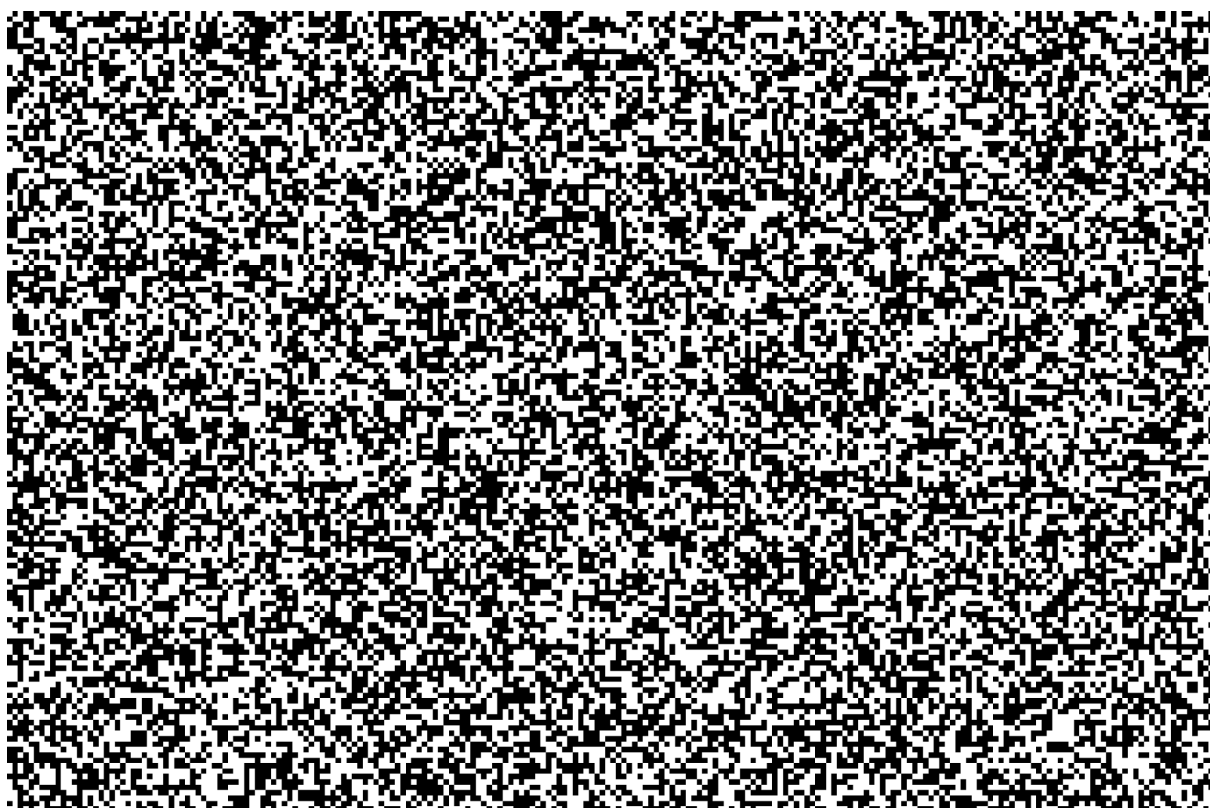
For architecture 2, the initial image had a penalty score of 155855.5000, the optimized image gave penalty score of -1058.8170 with $E/E_s = 0.0045$ and $v = -2.6149$.

Experiment 4: Fig. 5.7 (a) shows the initial image and Fig. 5.7 (b) shows the optimized image from architecture 1 and Fig. 5.7 (c) shows the optimized image from architecture 2.

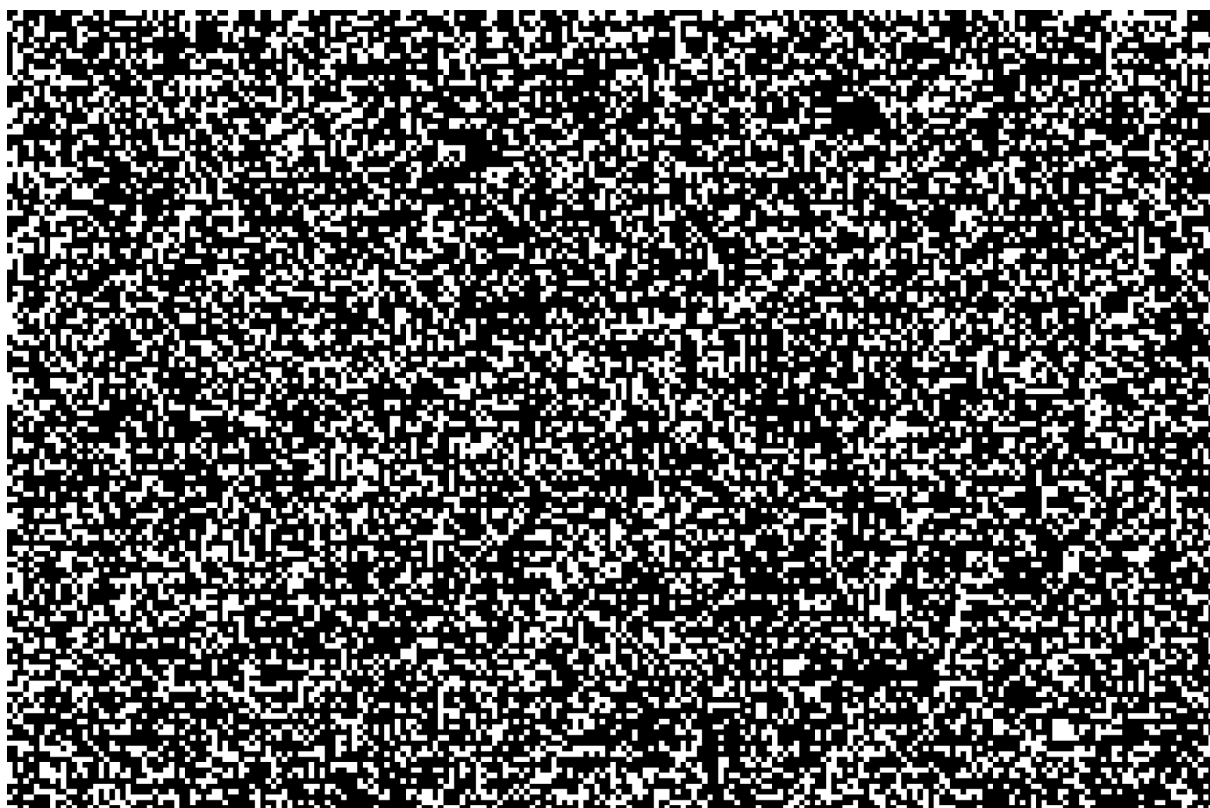
Here we start with a completely filled image and this time let it run for 1000 iterations instead of 100. The initial image is bordered for clarity.

For architecture 1, the initial image had a penalty score of 78784.0781 the optimized image gave penalty score of -1904.7563 with $E/E_s = 0.0019$ and $v = -4.7475$.

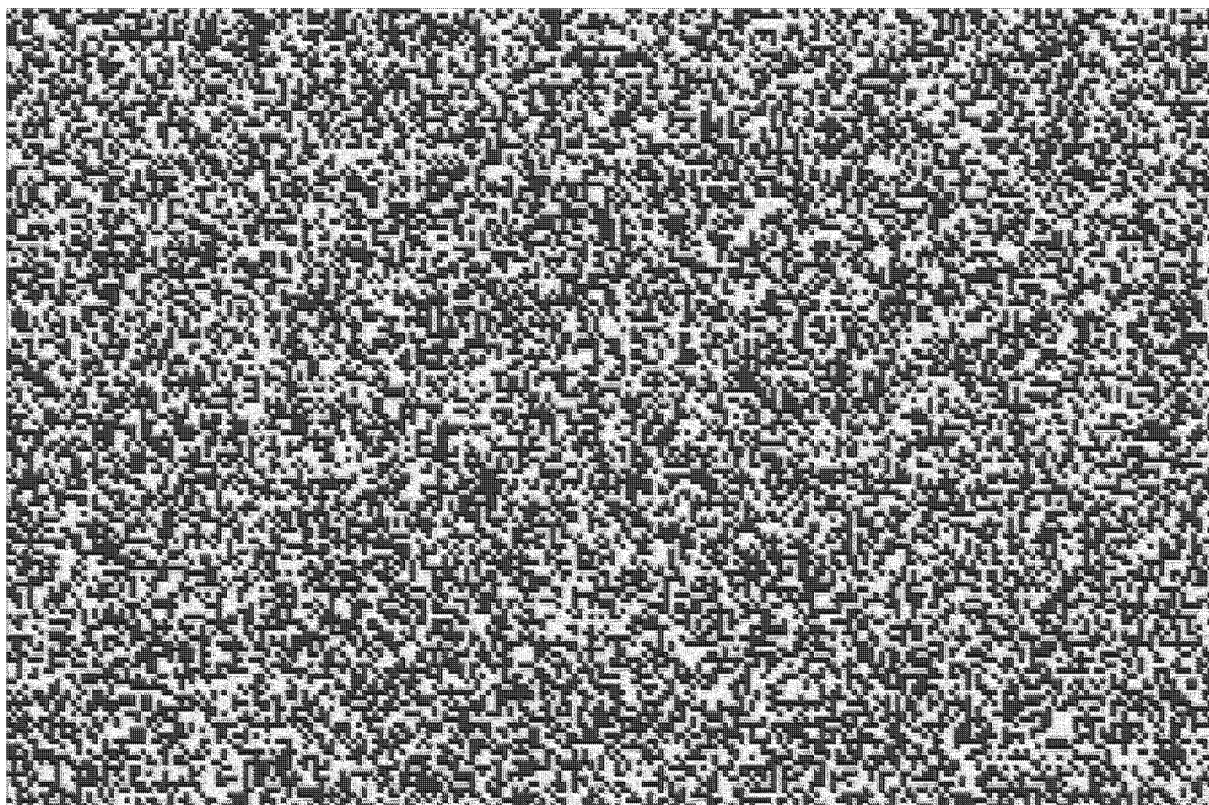
For architecture 2, the initial image had a penalty score of 219633.9531, the optimized image gave penalty score of -2584.5825 with $E/E_s = 0.0002$ and $v = -6.4601$.



(a)

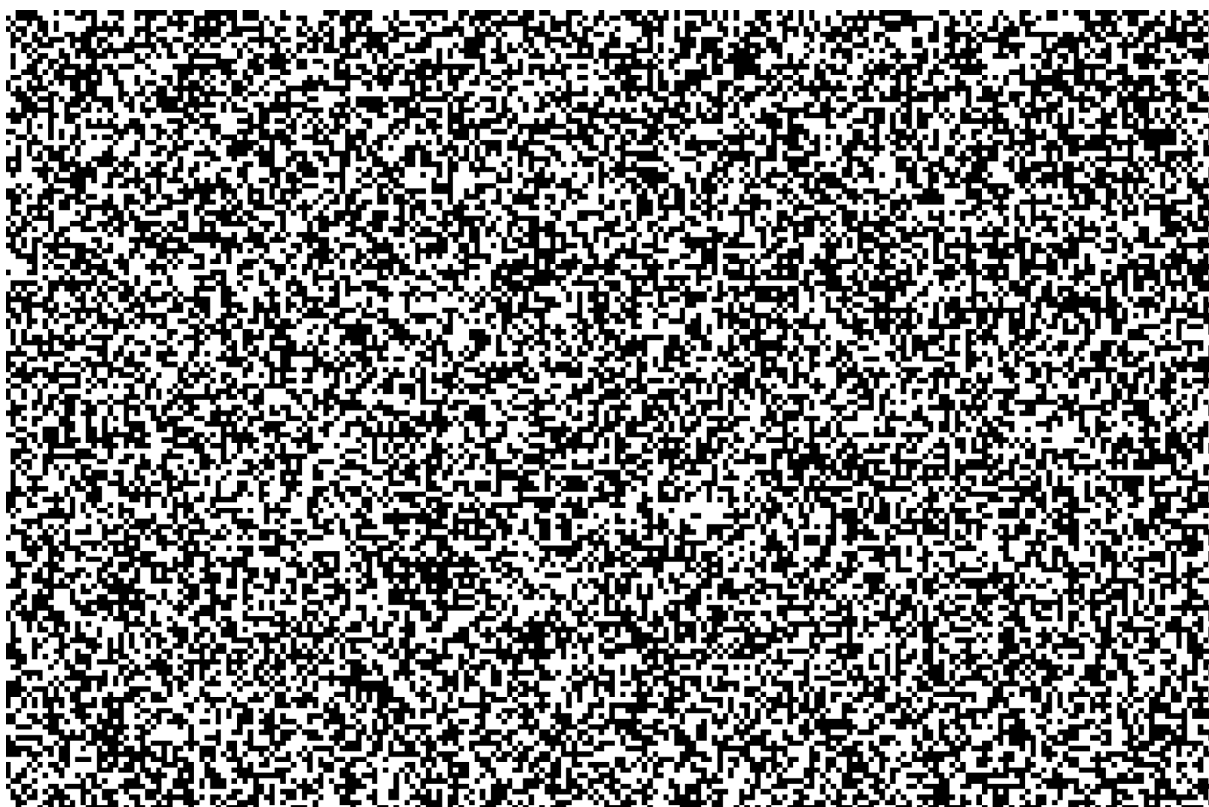


(b)

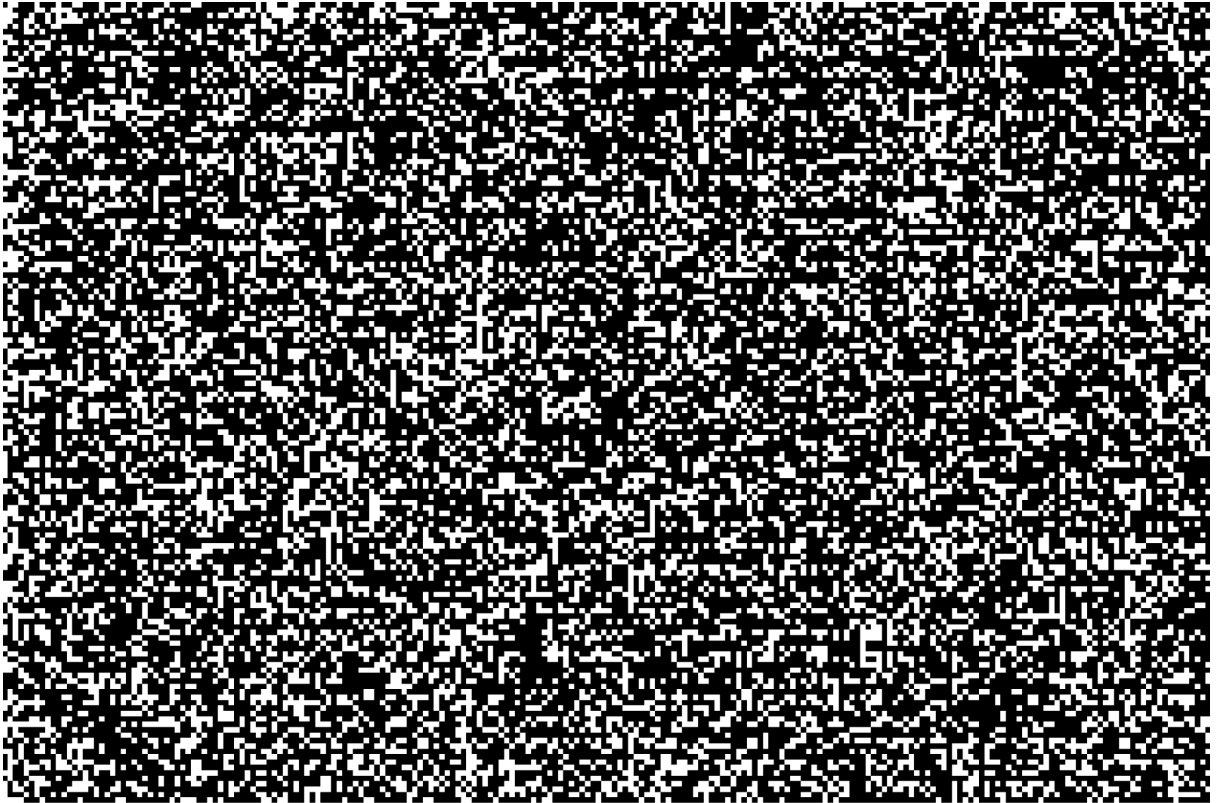


(c)

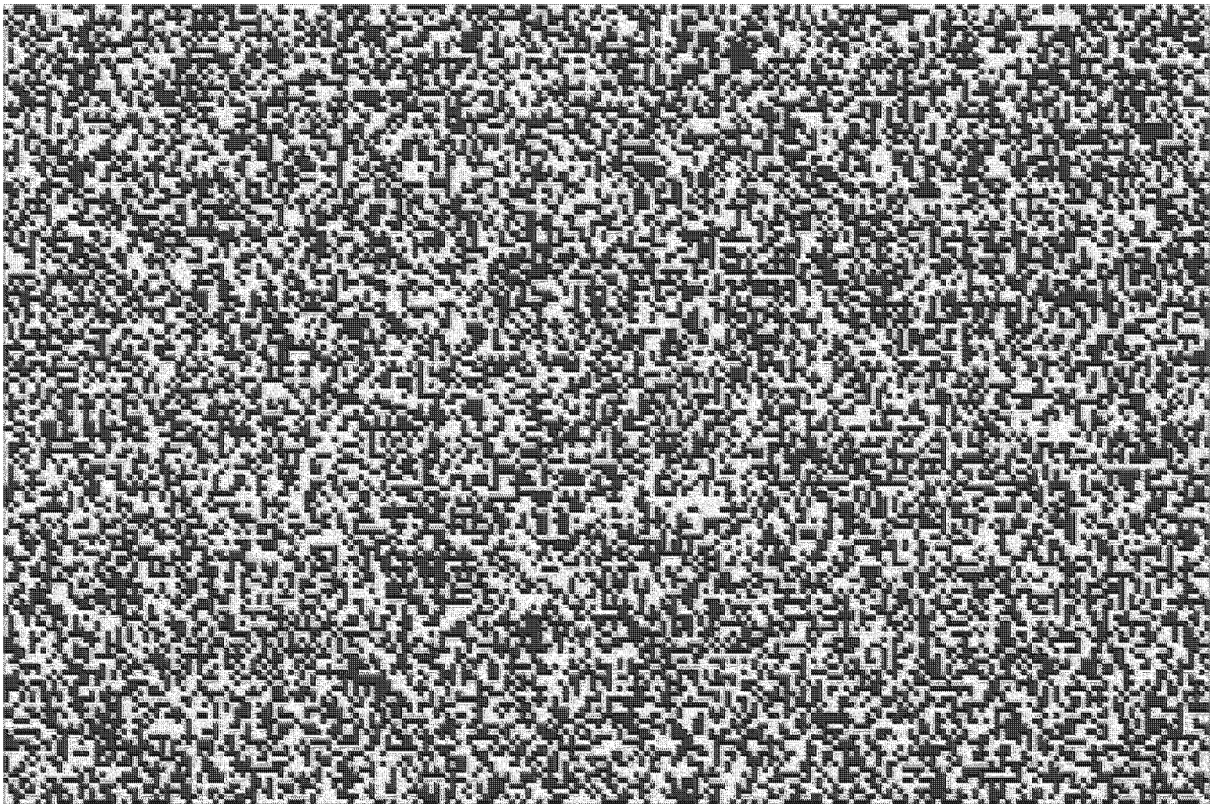
Fig. 5.4 (a) Initial reinforcement, Final Optimized Image for architecture (b) 1, (c) 2



(a)

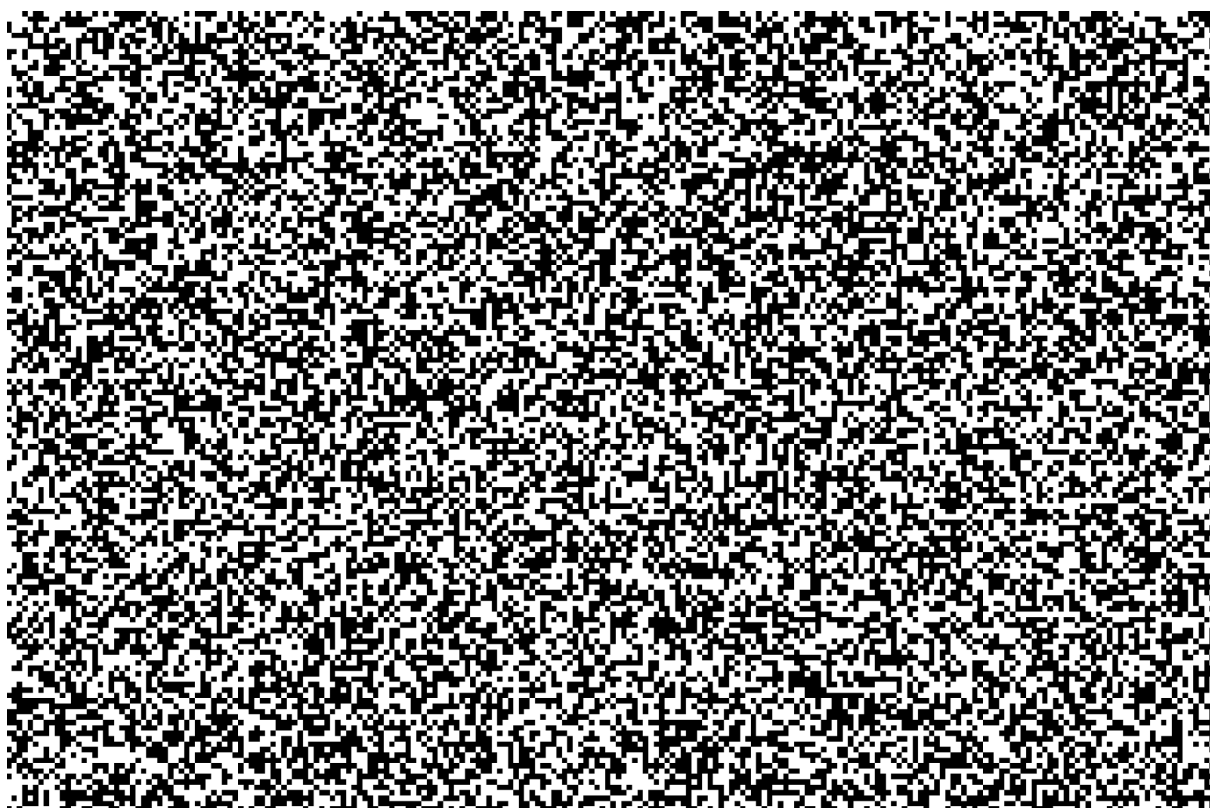


(b)

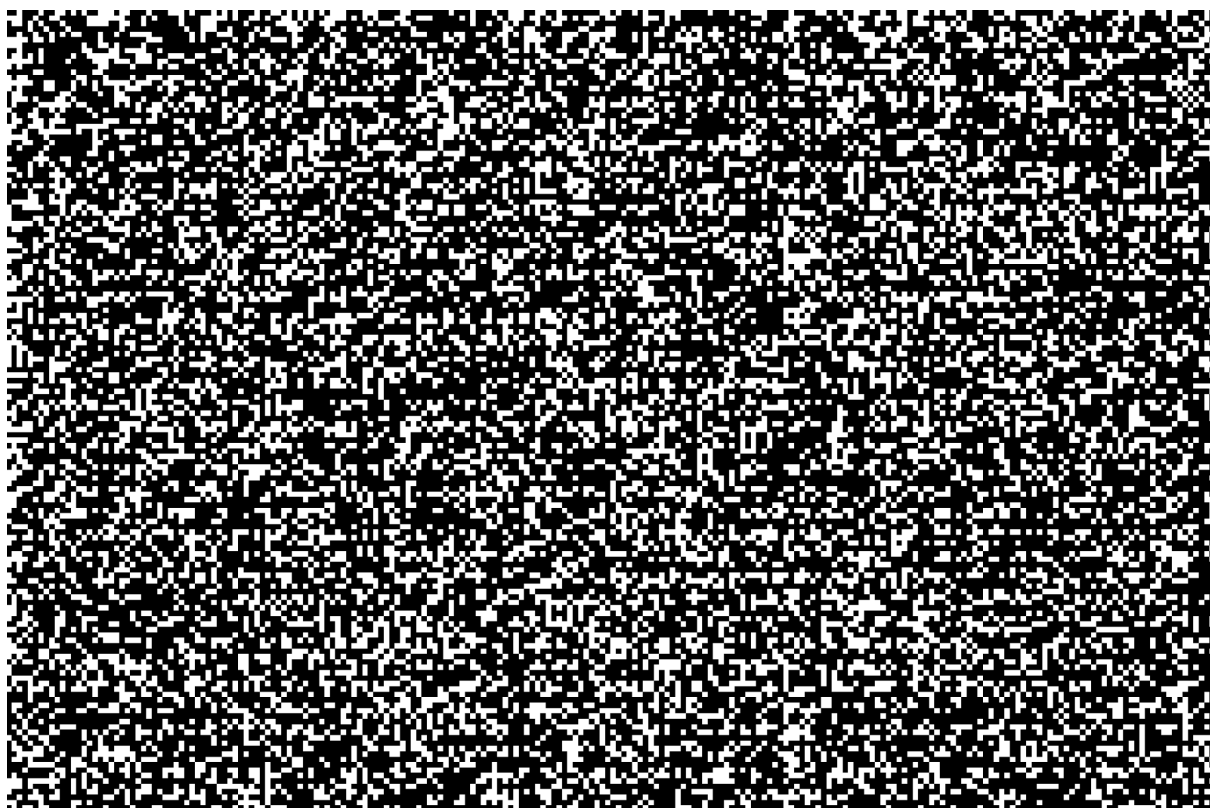


(c)

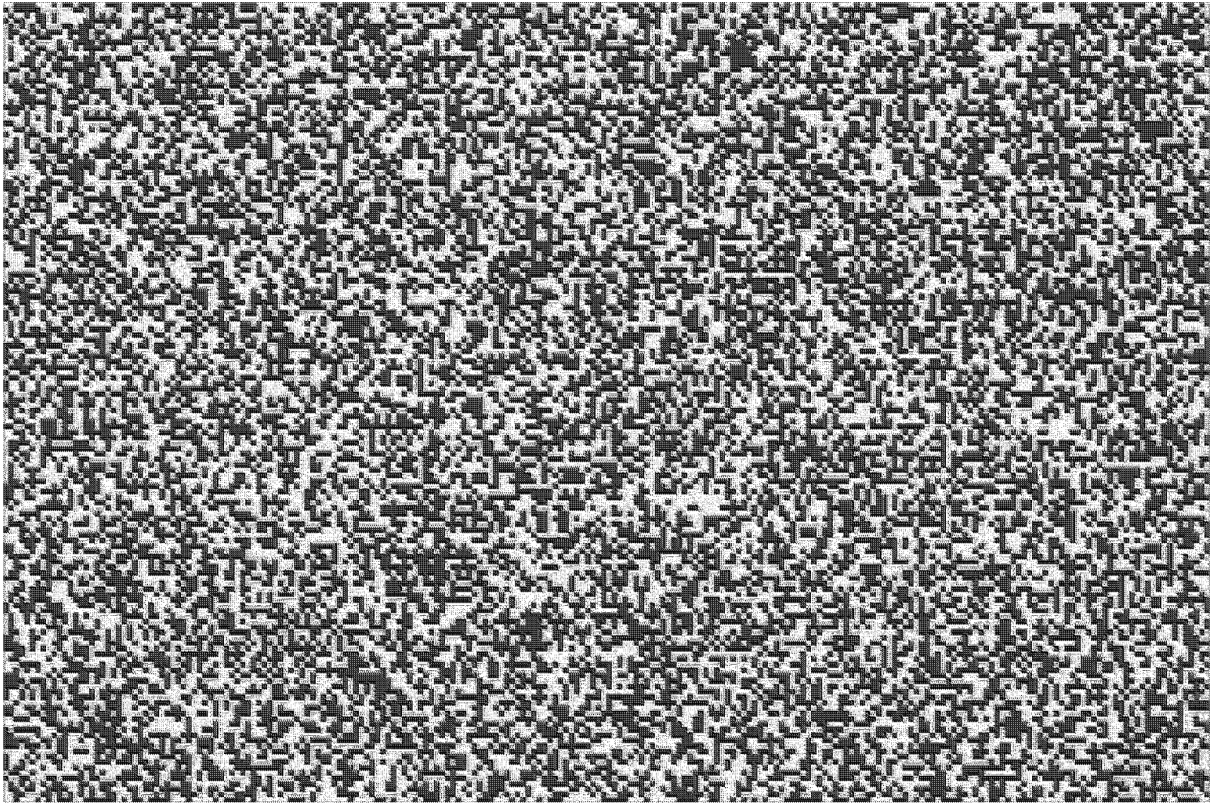
Fig. 5.5 (a) Initial reinforcement, Final Optimized Image for architecture (b) 1, (c) 2



(a)

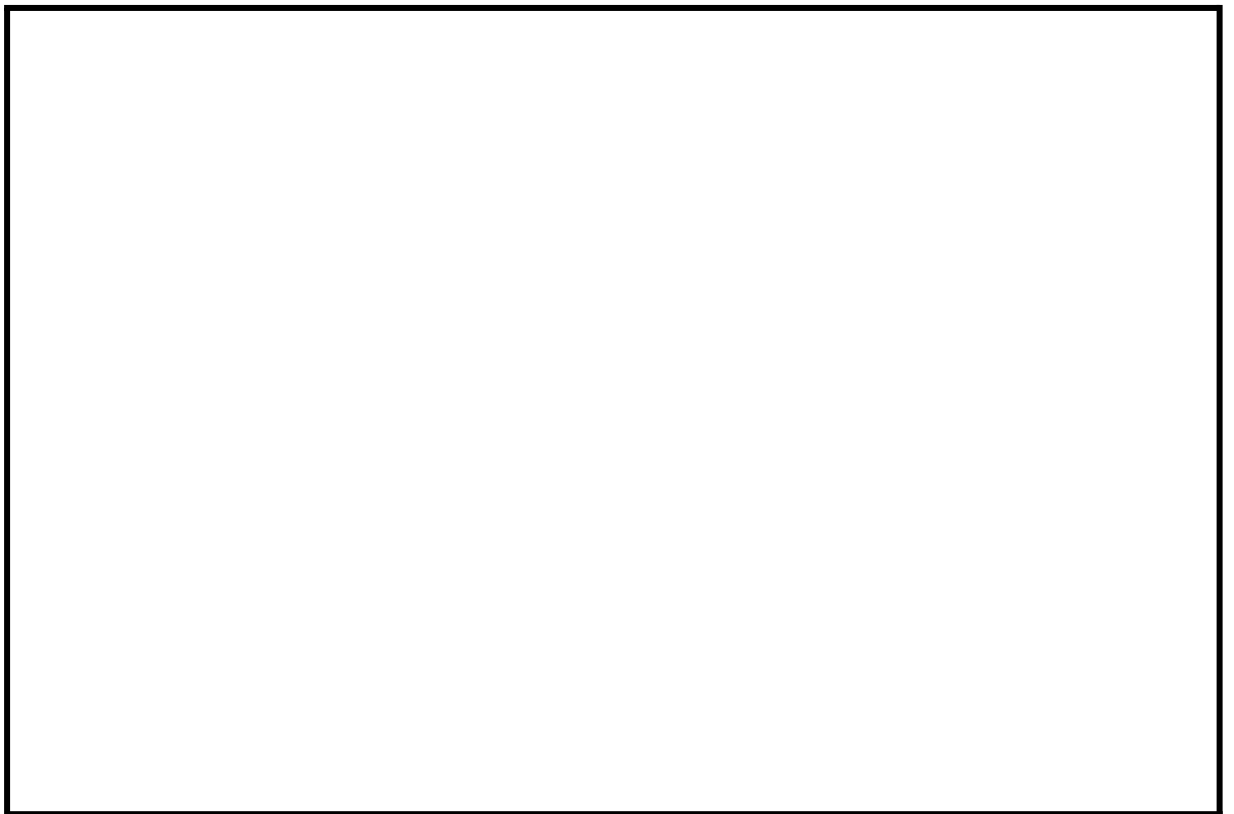


(b)



(c)

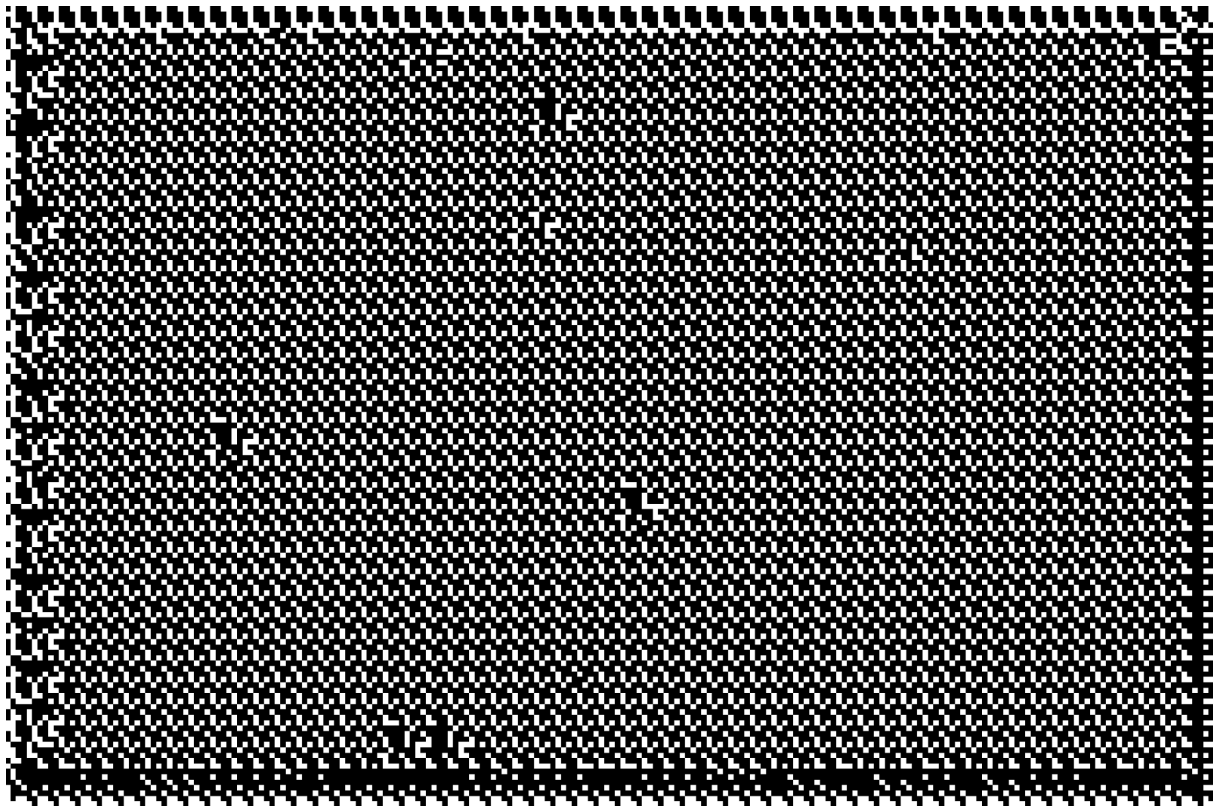
Fig. 5.6 (a) Initial reinforcement, Final Optimized Image for architecture (b) 1, (c) 2



(a)



(b)



(c)

Fig. 5.7 (a) Initial reinforcement, Final Optimized Image for architecture (b) 1, (c) 2

Those seem unrealistic numbers but this is when we pushed the optimization a lot and it kept on going to optimize for better v . This experiment produces some promising patterns as there was a lack of noise in the initial image. These can give some promising global patterns with qualitative appeal.

5.4 Post Processing Optimized Image Output

The optimized reinforcement image obtained through gradient-based input optimization initially exhibited high-frequency pixel noise with no immediately interpretable structure. However, upon applying the density-based post-processing pipeline described previously, coherent and spatially continuous reinforcement regions emerged.

The resulting processed image reveals the formation of distinct reinforcement zones characterized by clustered high-density regions. These regions are separated by relatively sparse areas, indicating that the optimization process has implicitly learned a non-uniform distribution of reinforcement rather than a trivial uniform solution.

The result obtained in Fig. 5.7 (b) from architecture 1 shows a promising structure with well defined region boundaries and a potential to produce a physically meaningful reinforcement.

Using the simple post processing method discussed in section 4.11, the resulting processed image can be produced as shown in Fig. 5.8.

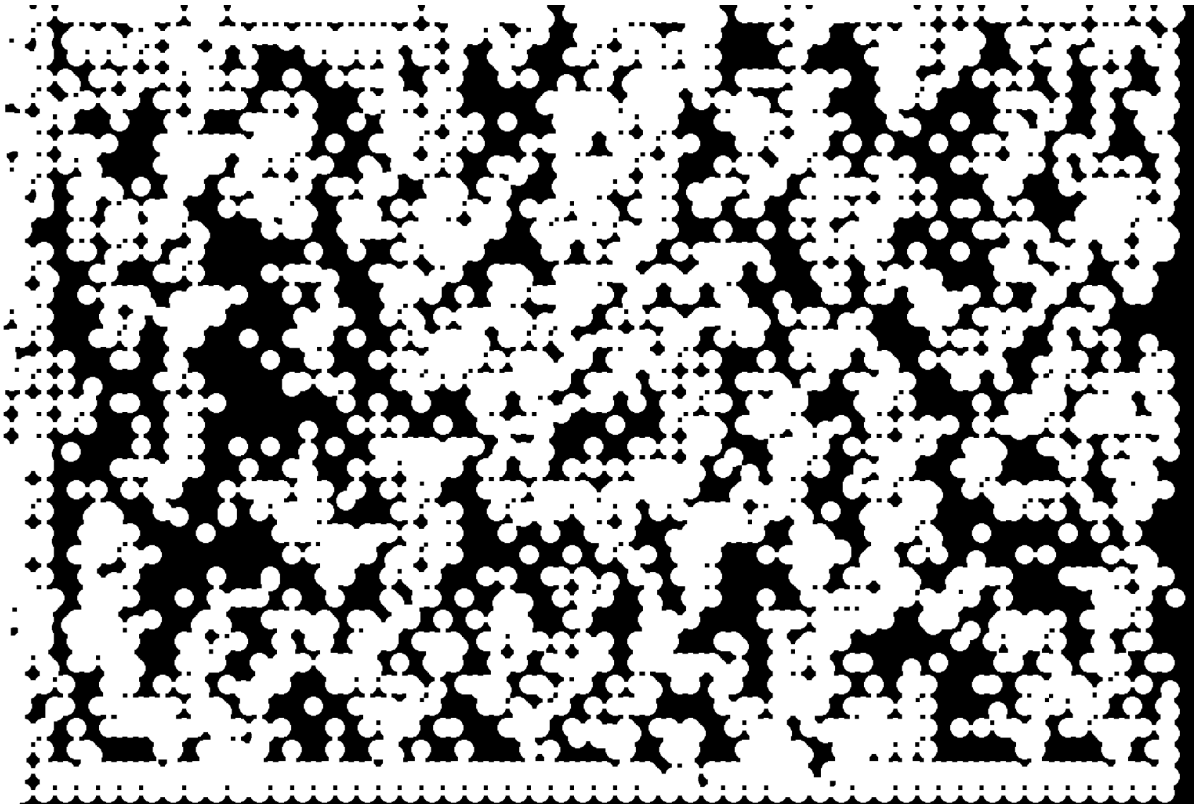


Fig. 5.8 Post processed output for the obtained reinforcement from Architecture 1

The image generated as the result for post-processing shown in Fig. 5.8 is fed back to the trained Architecture 1 model and the predicted values of $E/E_s = 0.1057$ and $\nu = -0.7741$.

More sophisticated methods for image pre-processing may be employed as well to have the region boundaries as smooth as possible. It should be ensured in all those methods that there is no independent region in that reinforcement that is small enough to collapse the lattice.

6. Conclusions and Future Work

This work presents a complete data-driven framework for the analysis and optimization of reinforced re-entrant honeycomb (REH) lattices by integrating finite element simulations with deep learning–based surrogate modeling and gradient-based inverse design.

A large-scale dataset (~9600 samples, as shown in Section 5.1) was generated using an automated Abaqus-based pipeline, capturing the nonlinear relationship between reinforcement configurations and macroscopic mechanical properties. The results clearly establish the inherent trade-off: increasing reinforcement improves stiffness (E/E_s) but simultaneously reduces auxeticity (makes Poisson’s ratio less negative).

The trained convolutional neural network successfully learned this complex mapping, achieving high predictive accuracy ($R^2 \approx 0.95$ – 0.96 across both properties). Notably, even relatively small architectures were sufficient, indicating that the reinforcement–property relationship is structured and learnable through spatial patterns rather than requiring excessively deep models.

The trained model was then utilized as a differentiable surrogate to perform gradient-based optimization directly in the image space of reinforcement patterns. This demonstrated the feasibility of inverse design without repeated finite element simulations, significantly reducing computational cost. The optimization process consistently converged to configurations that improved the defined objective function, validating the effectiveness of the surrogate-driven approach.

However, the raw optimized outputs revealed a key limitation: direct pixel-wise optimization produces high-frequency, non-physical patterns. This necessitated a post-processing pipeline based on density estimation and morphological operations, which successfully transformed noisy outputs into structured, interpretable reinforcement regions.

The study highlights several important insights:

- Reinforcement effectiveness is governed by spatial distribution and connectivity rather than mere quantity
- There exists a strong and unavoidable trade-off between stiffness enhancement and auxetic behavior
- Surrogate-based optimization can uncover non-trivial reinforcement patterns beyond manual design intuition
- Post-processing is essential to bridge the gap between optimization space and physically realizable structures

Despite these promising results, limitations remain. The optimization is performed in a relaxed continuous image space and does not inherently enforce manufacturability or strict geometric constraints. Additionally, the optimized configurations require validation through finite element simulations to confirm their physical accuracy.

Future work should focus on incorporating geometric constraints directly into the optimization process, improving regularization to reduce noise generation, and scaling the dataset to enhance model generalization. Extending this framework to 3D structures and integrating physics-informed learning could further improve robustness and applicability.

In conclusion, this work demonstrates that combining finite element simulations with deep learning and gradient-based optimization provides a powerful and efficient framework for the inverse design of auxetic metamaterials, enabling exploration of complex design spaces that are otherwise computationally prohibitive.

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8. Appendix

The code for the entire pipeline can be found in the open source github repository linked to https://github.com/anscian/REH_reinforcement_optimization

The link to the dataset that was collected from multiple FE simulations is present in the README file for this repository.